

## Reminders

Upcoming due dates

This Fri D4

Next Wed A2

Proposal feedback in Github Issues by next Wednesday

# Linear models

**Data Science in Practice**

**TODO next quarter: check in after OLS**

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## CORRELATION

ASSOCIATION  
BETWEEN VARIABLES

i.e. Pearson  
Correlation,  
Spearman  
Correlation, chi-  
square test

## COMPARISON OF MEANS

DIFFERENCE IN MEANS  
BETWEEN VARIABLES

i.e. t-test, ANOVA

## REGRESSION

DOES CHANGE IN ONE  
VARIABLE MEAN CHANGE  
IN ANOTHER?

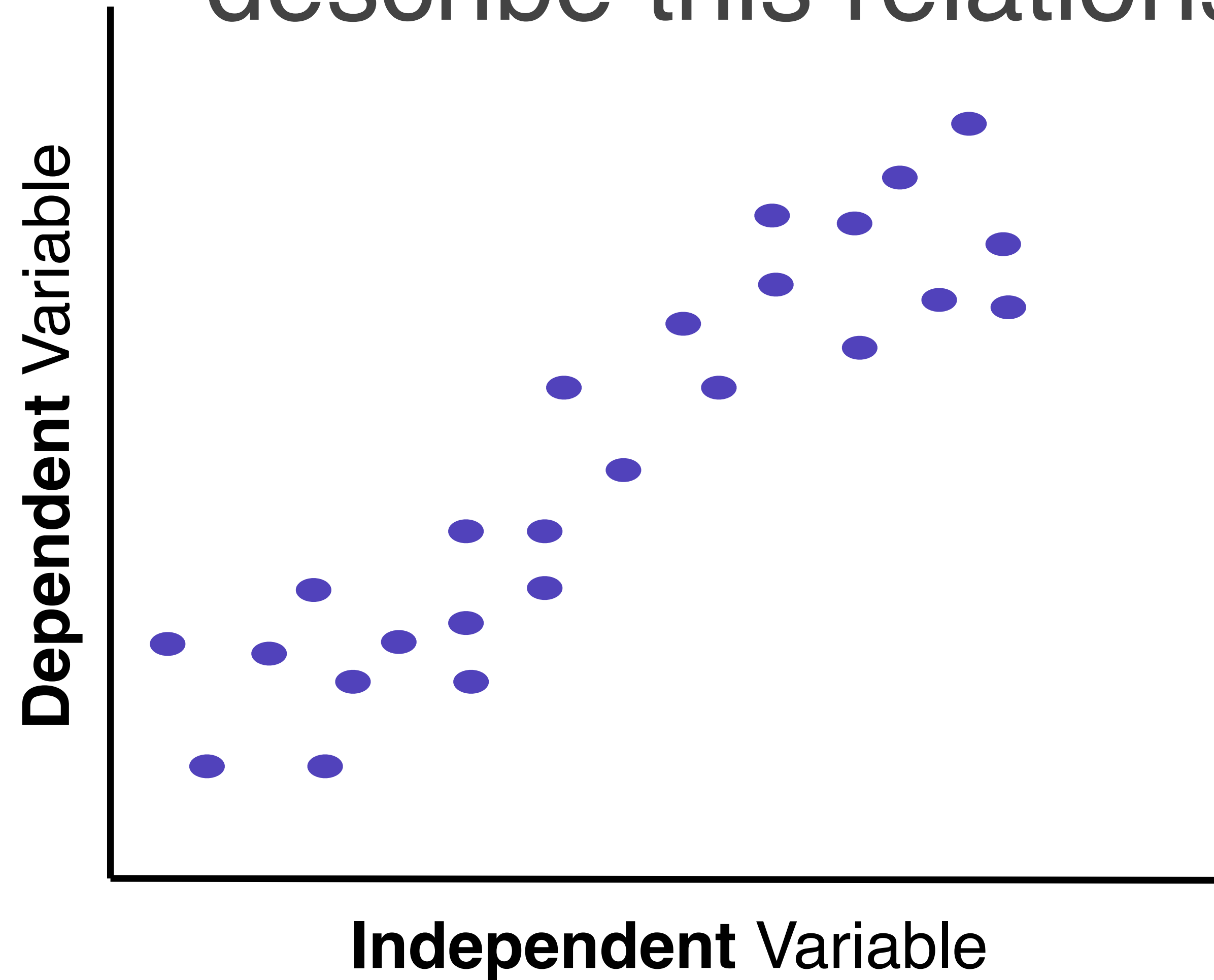
I.e. simple  
regression, multiple  
regression

## NON-PARAMETRIC TESTS

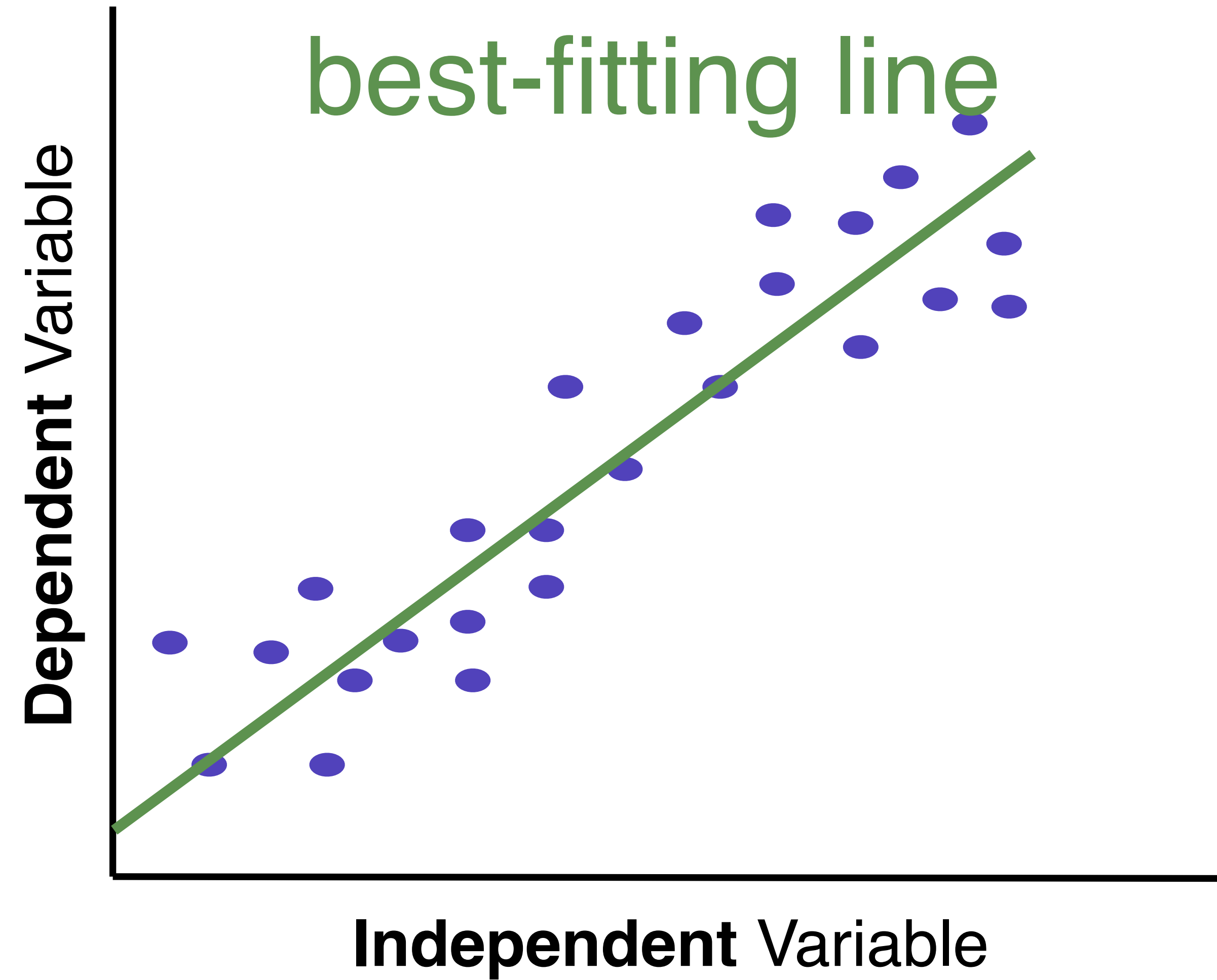
FOR WHEN ASSUMPTIONS  
IN THESE OTHER 3  
CATEGORIES ARE NOT  
MET

i.e. Wilcoxon rank-  
sum test, Wilcoxon  
sign-rank test, sign  
test

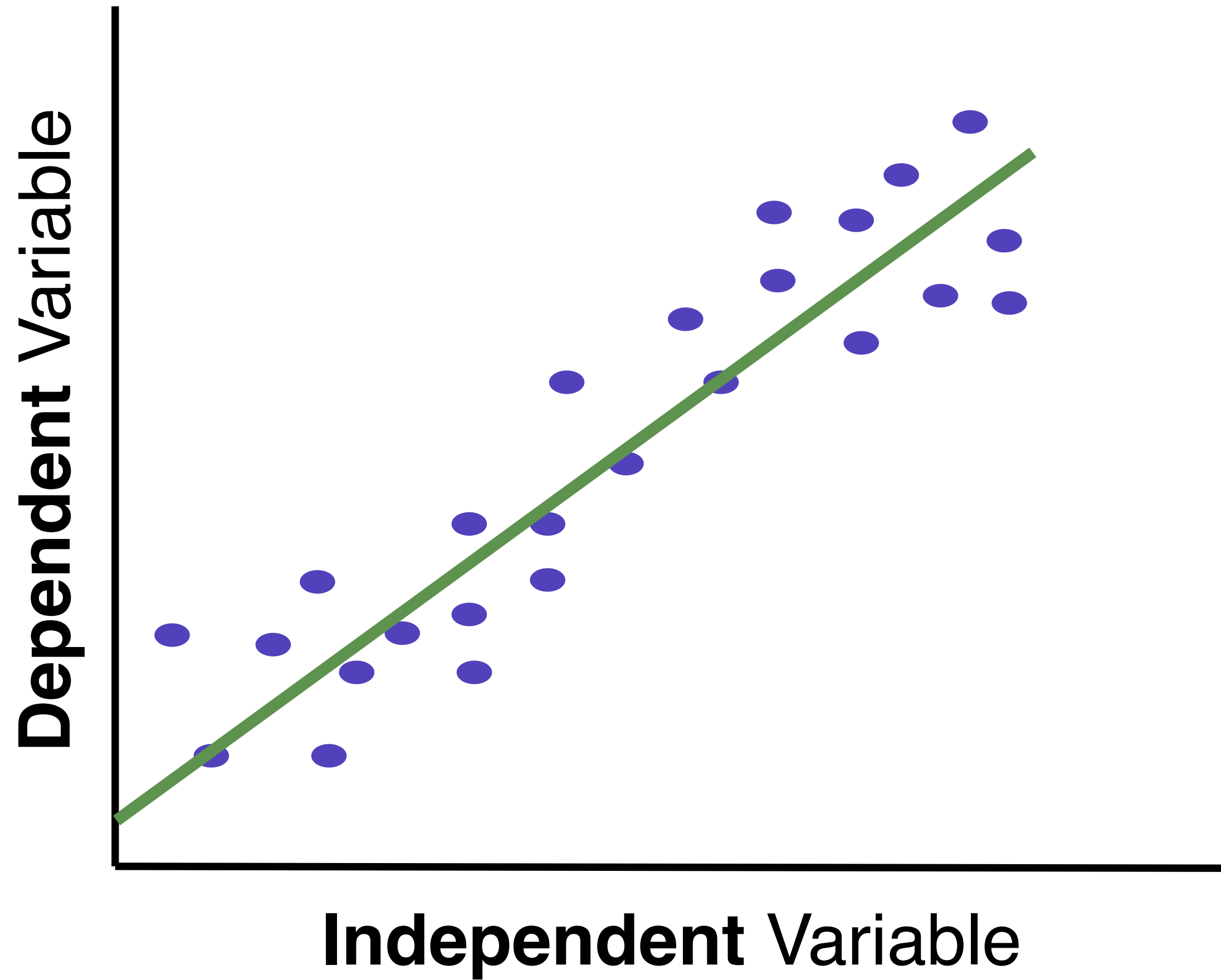
Linear regression can be used to describe this relationship



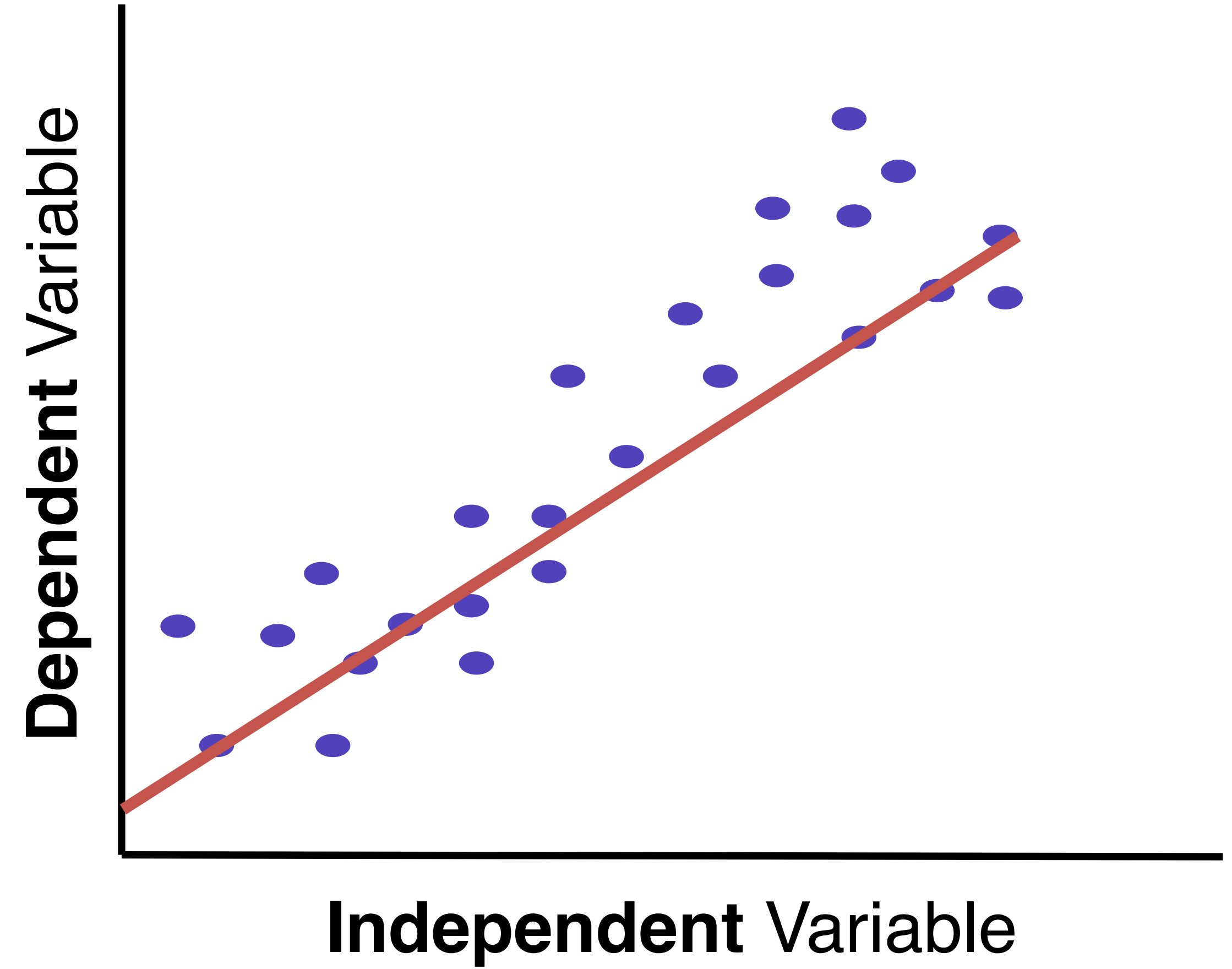




Best-fitting line



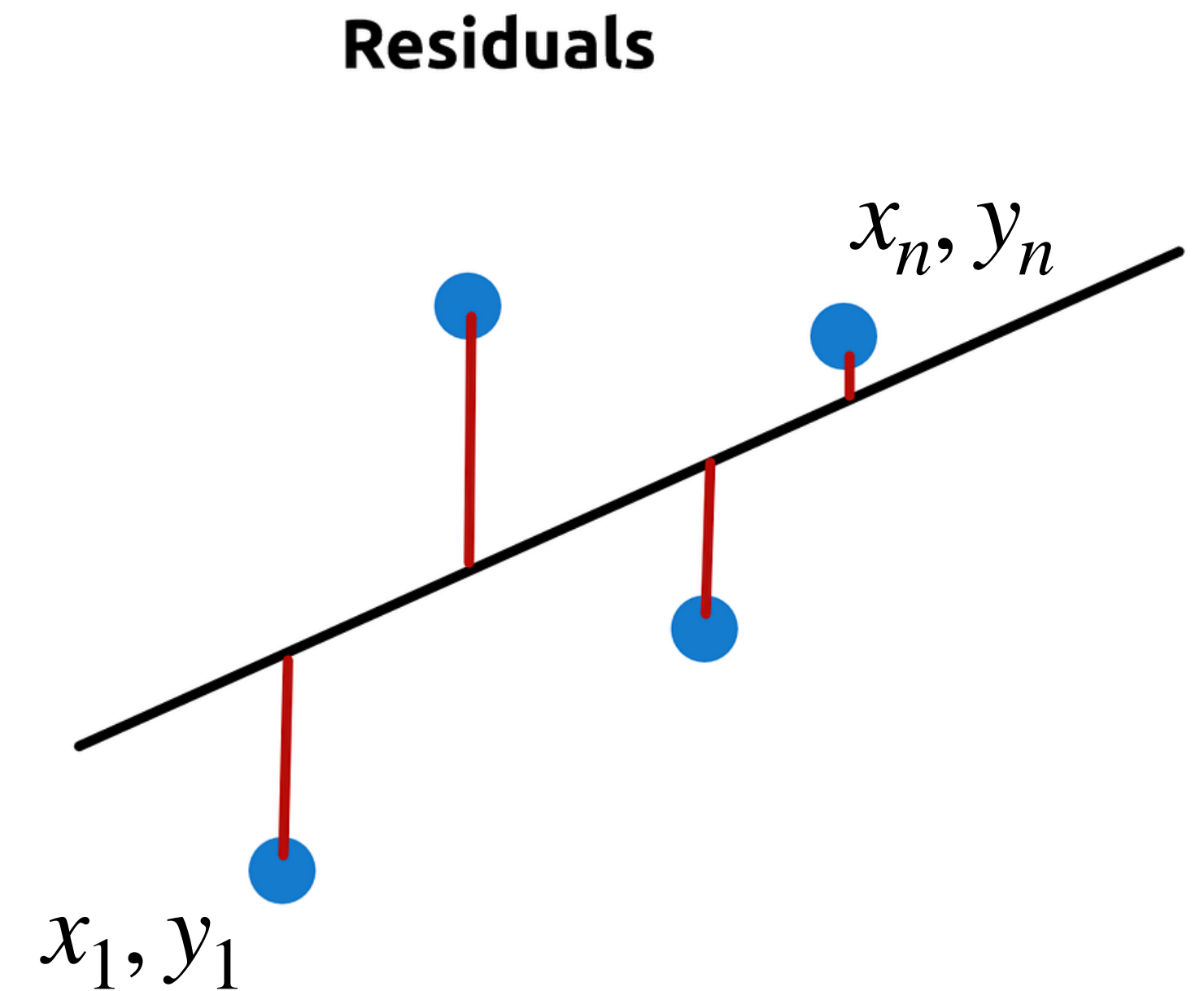
NOT a best-fitting line



# Ordinary Least Squares

Just one of several regression algorithms

- Minimizes Residual Sum of Squares (RSS)
- OLS residuals are VERTICAL only!!
  - Implies x predicts y values!
  - Compare with TLS regression which uses perpendicular distance
- OLS residuals are SQUARED!!
  - Outliers have large influence!
  - Compare with LAD (aka L1 loss) regression which uses absolute value instead of square

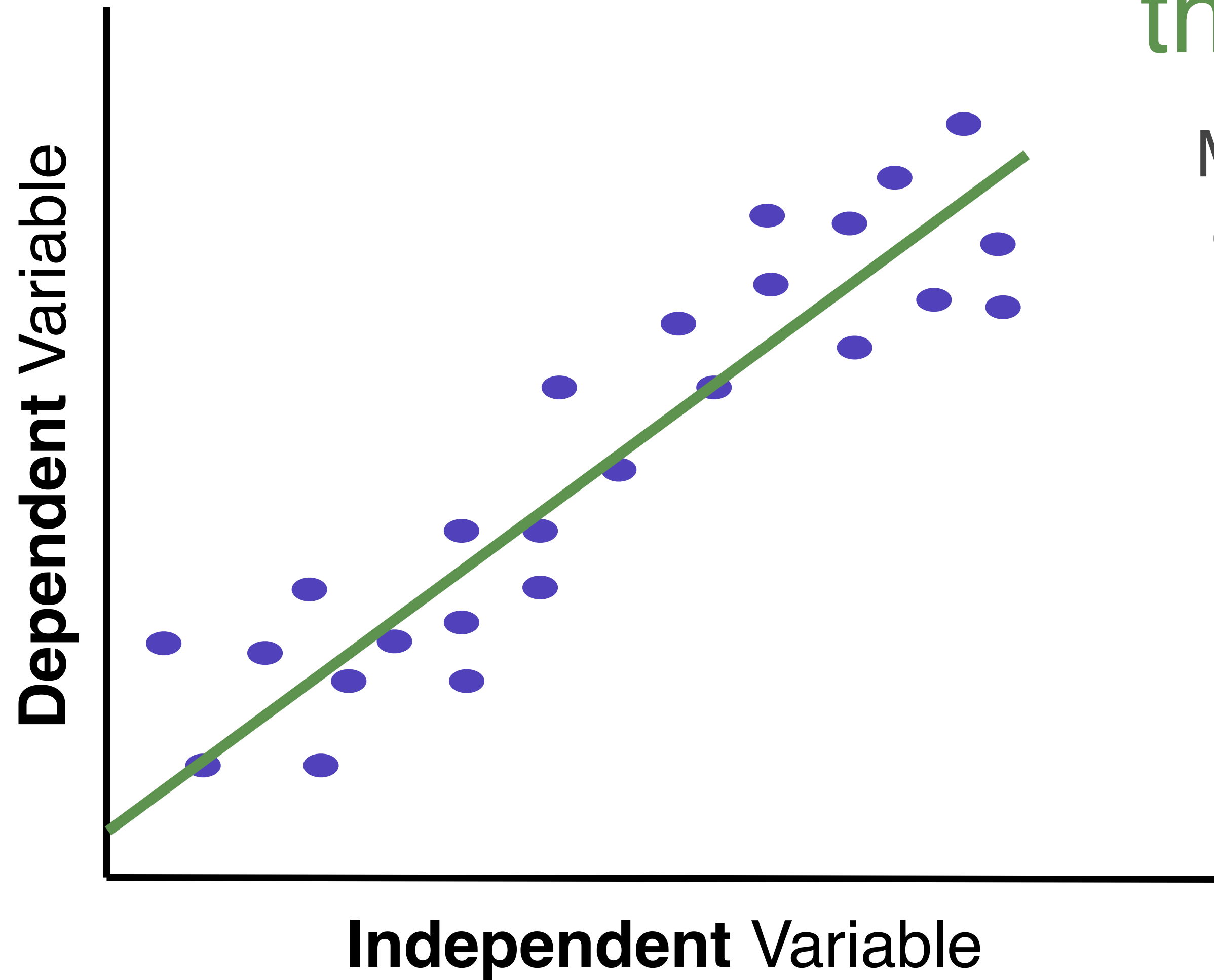


$$RSS = \sum_{k=1}^n (f(x_k) - y_k)^2$$



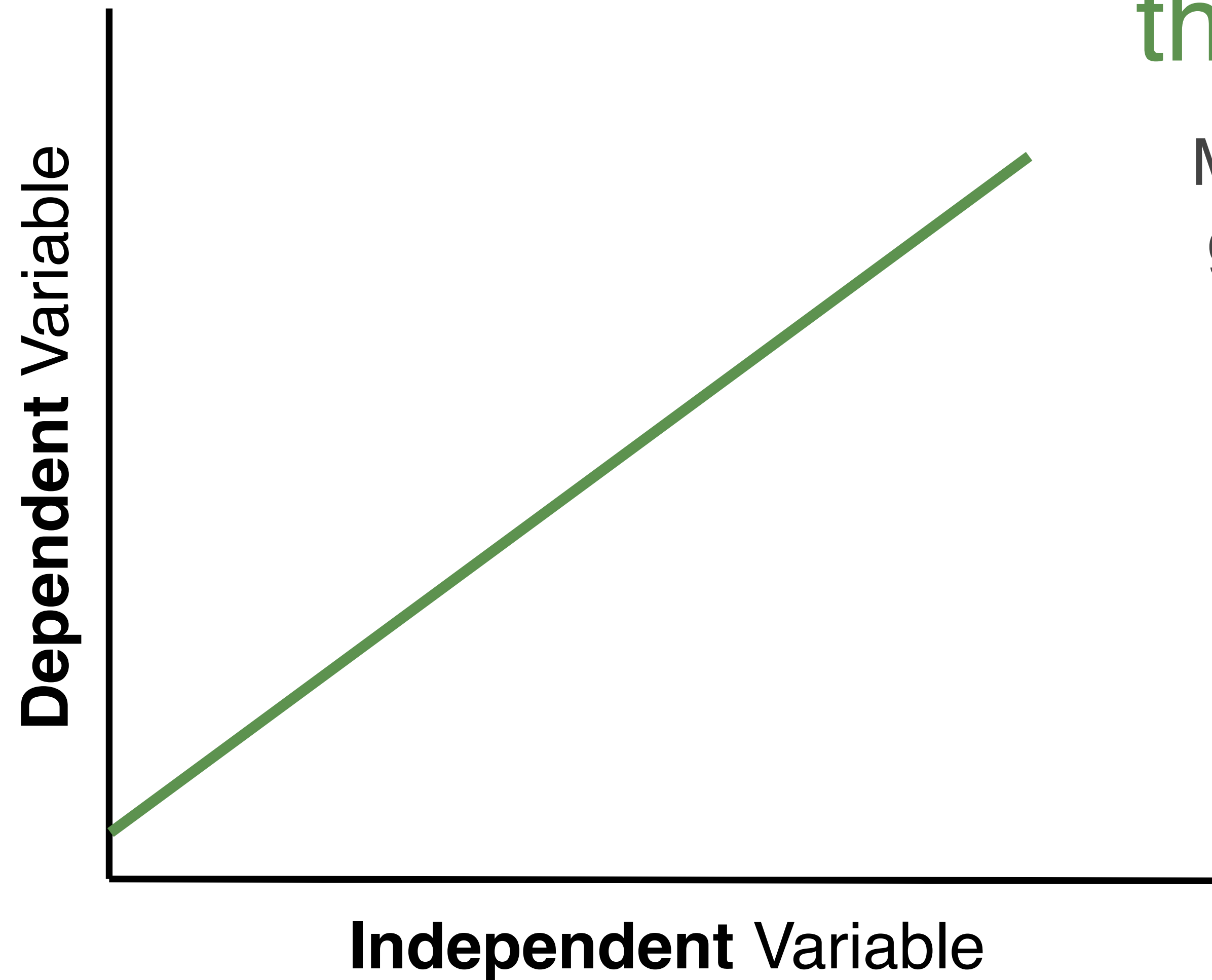
# This line is a model of the data

Models are mathematical equations generated to *represent* the real life situation



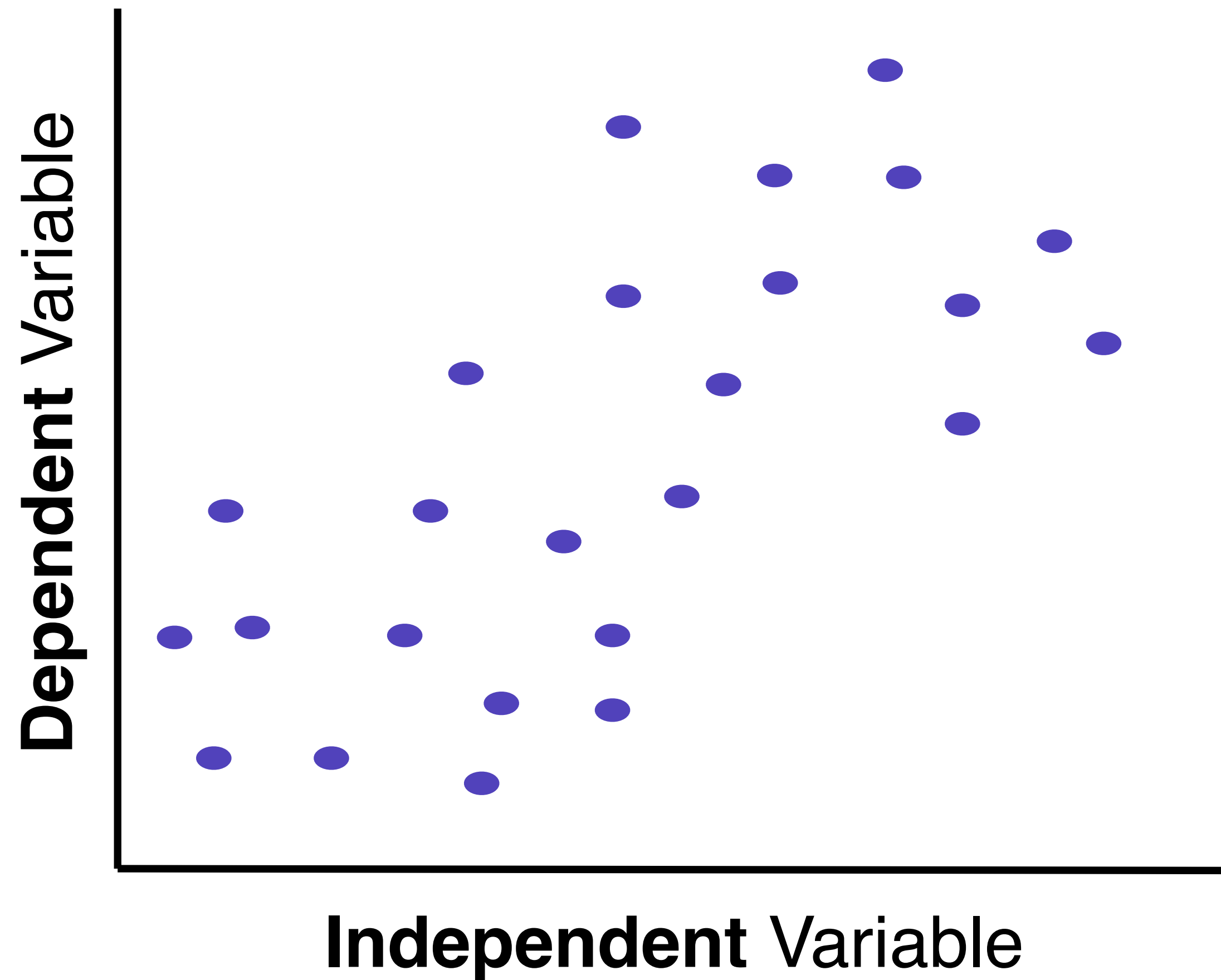
# This line is a model of the data

Models are mathematical equations generated to *represent* the real life situation

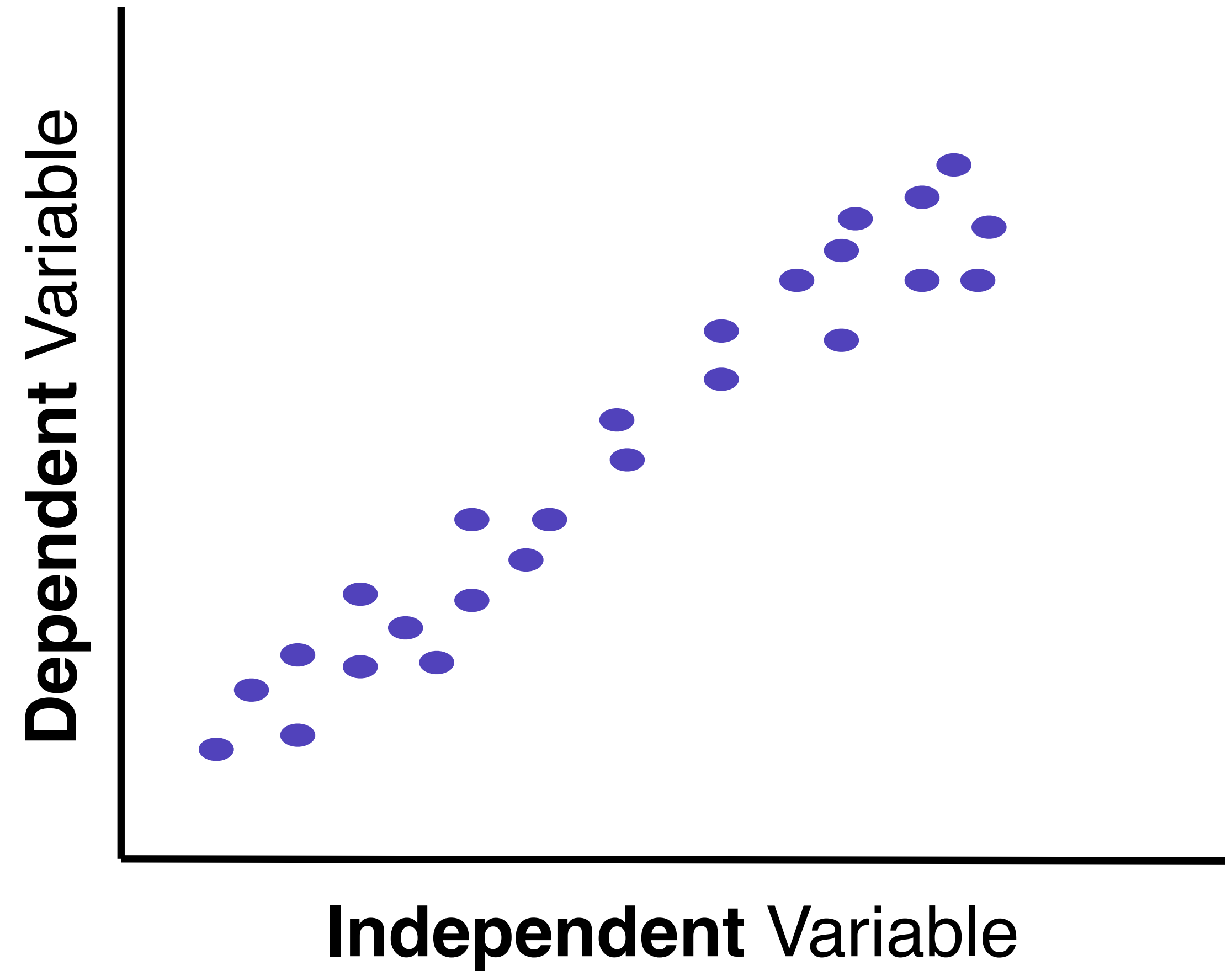




weaker relationship



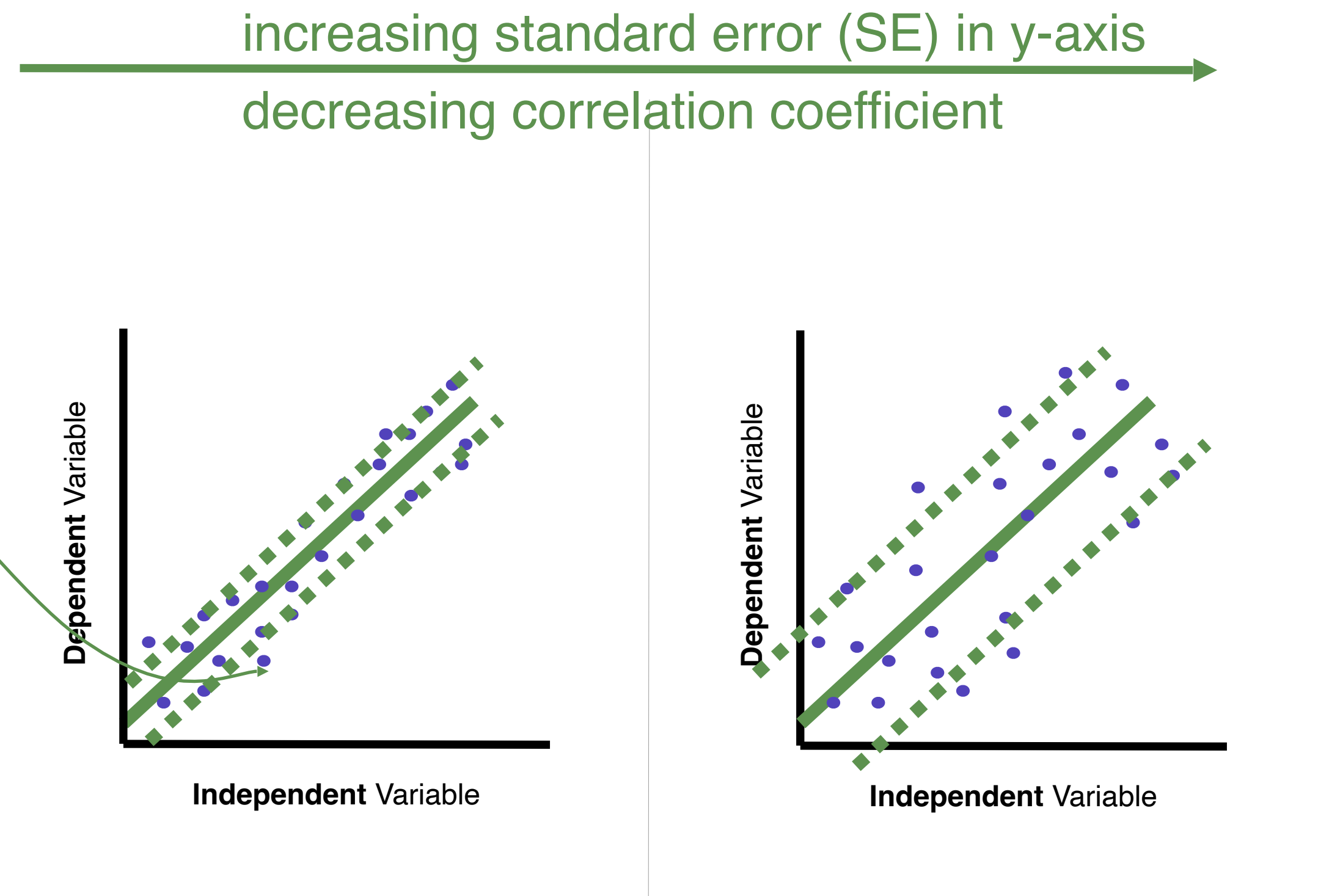
stronger relationship



stronger relationship = higher correlation

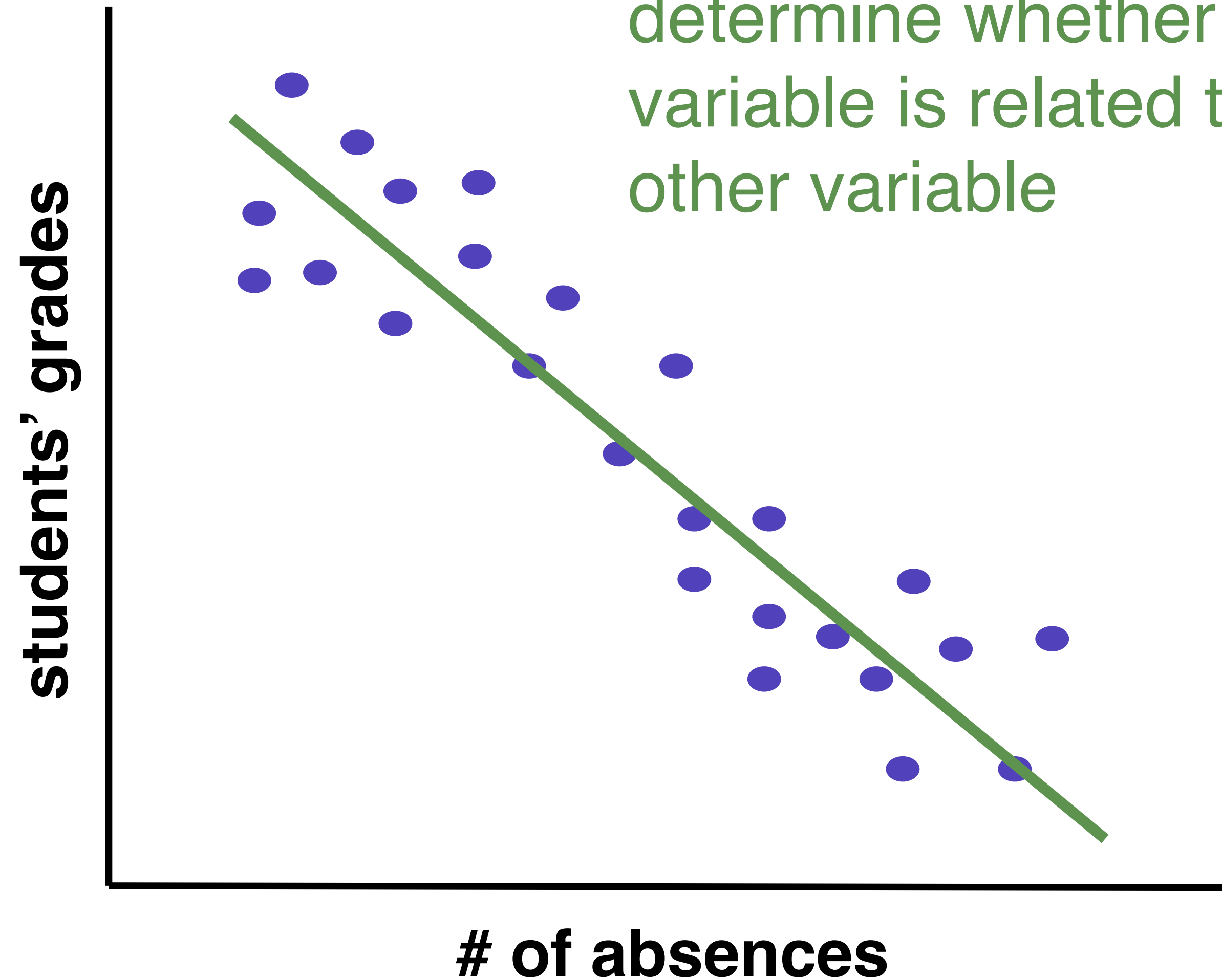
Pearson's R squared is Coefficient of Determination

The *closer* the points are to the regression line, the *less uncertain* we are in our estimate



Standard error is standard deviation /  $\sqrt{n}$   
Decreasing standard error means  
more deterministic relationship and  
more certainty in the regression

Linear regression can be used to determine whether a change in one variable is related to the change in the other variable





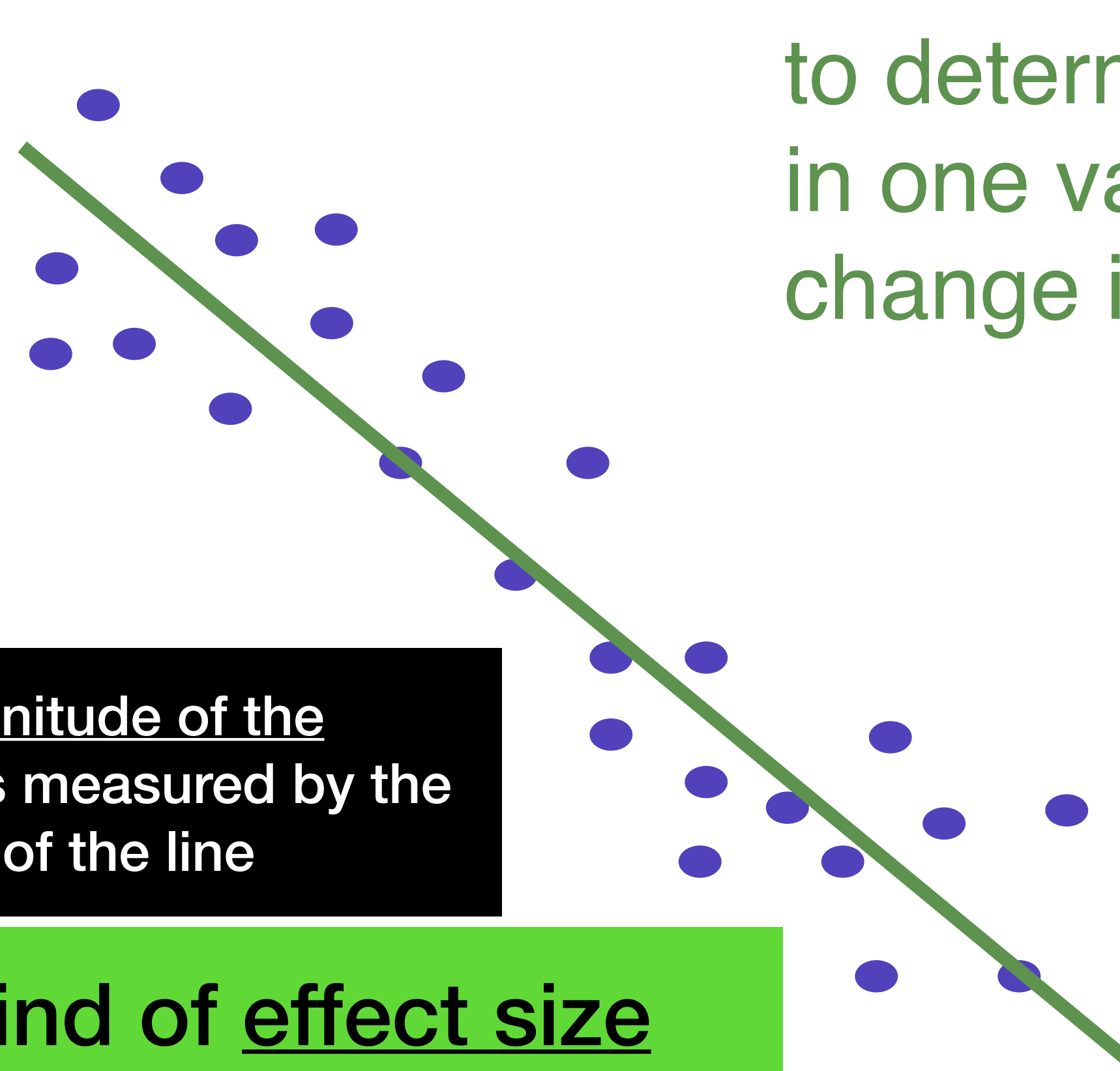
Linear regression can be used to determine whether a change in one variable is related to the change in the other variable

nts' grades

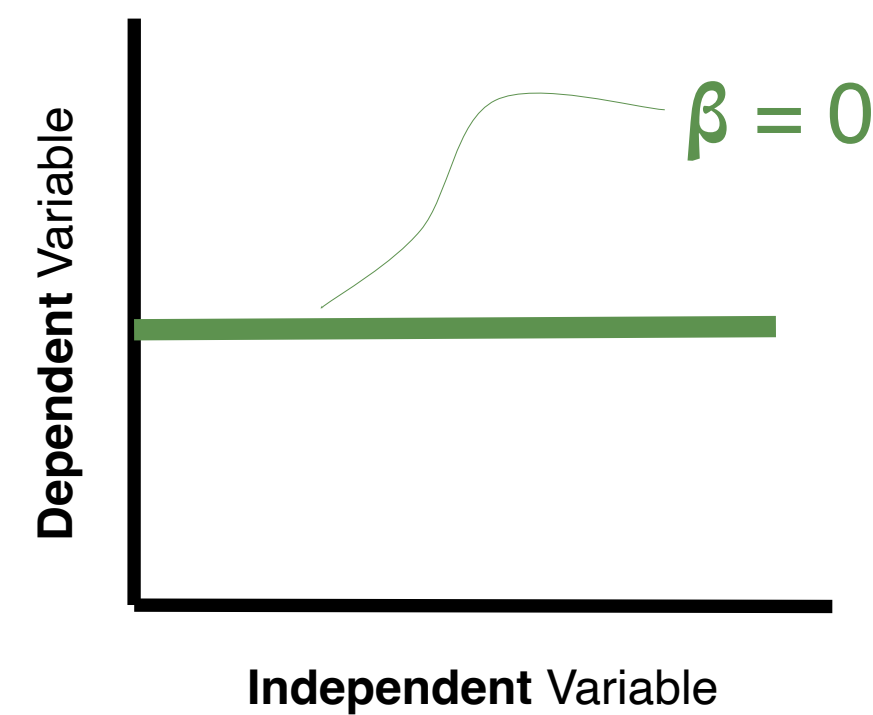
The magnitude of the relationship is measured by the slope of the line

This is another kind of effect size

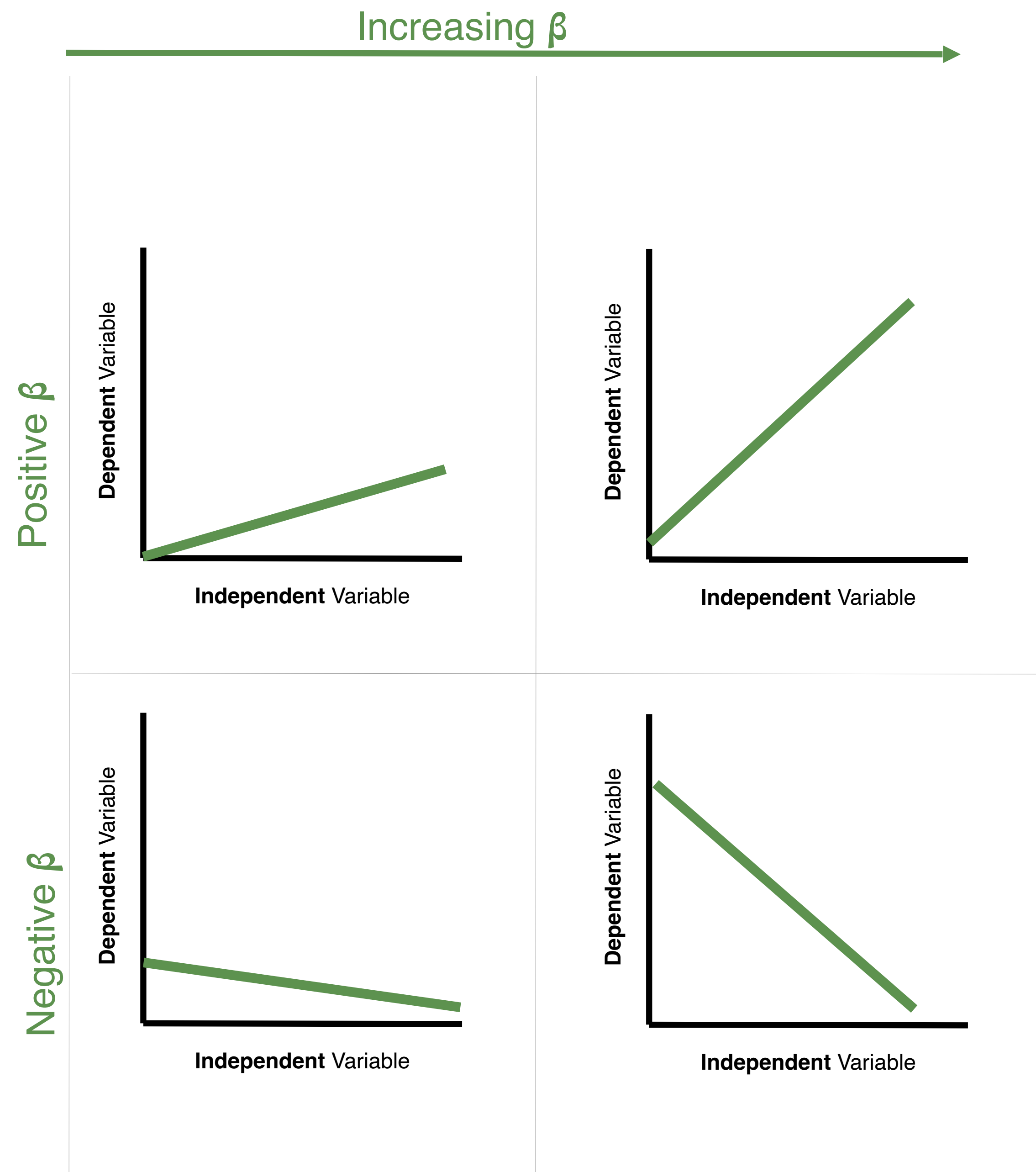
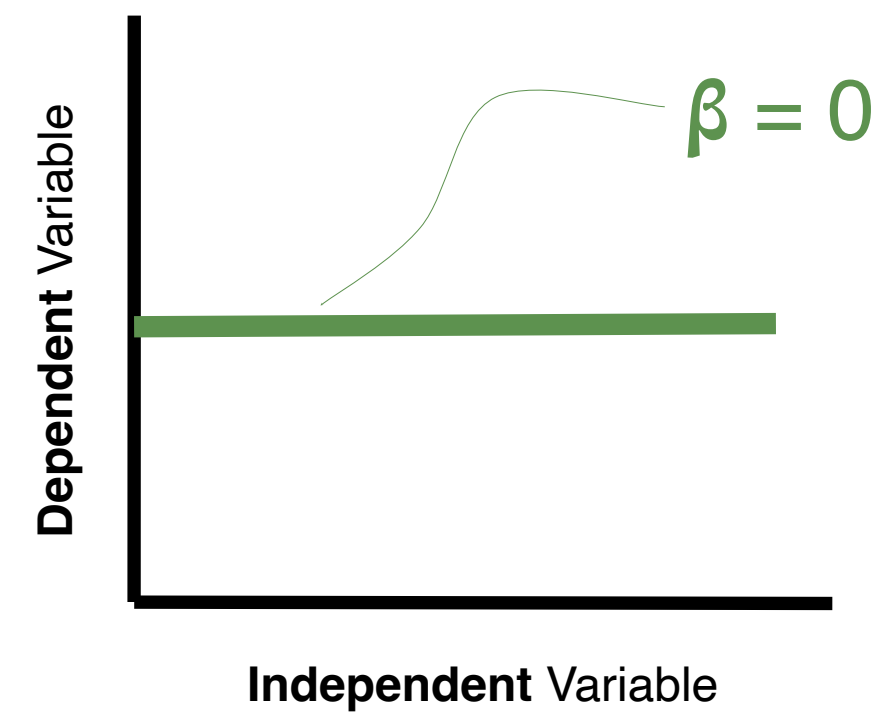
# of absences



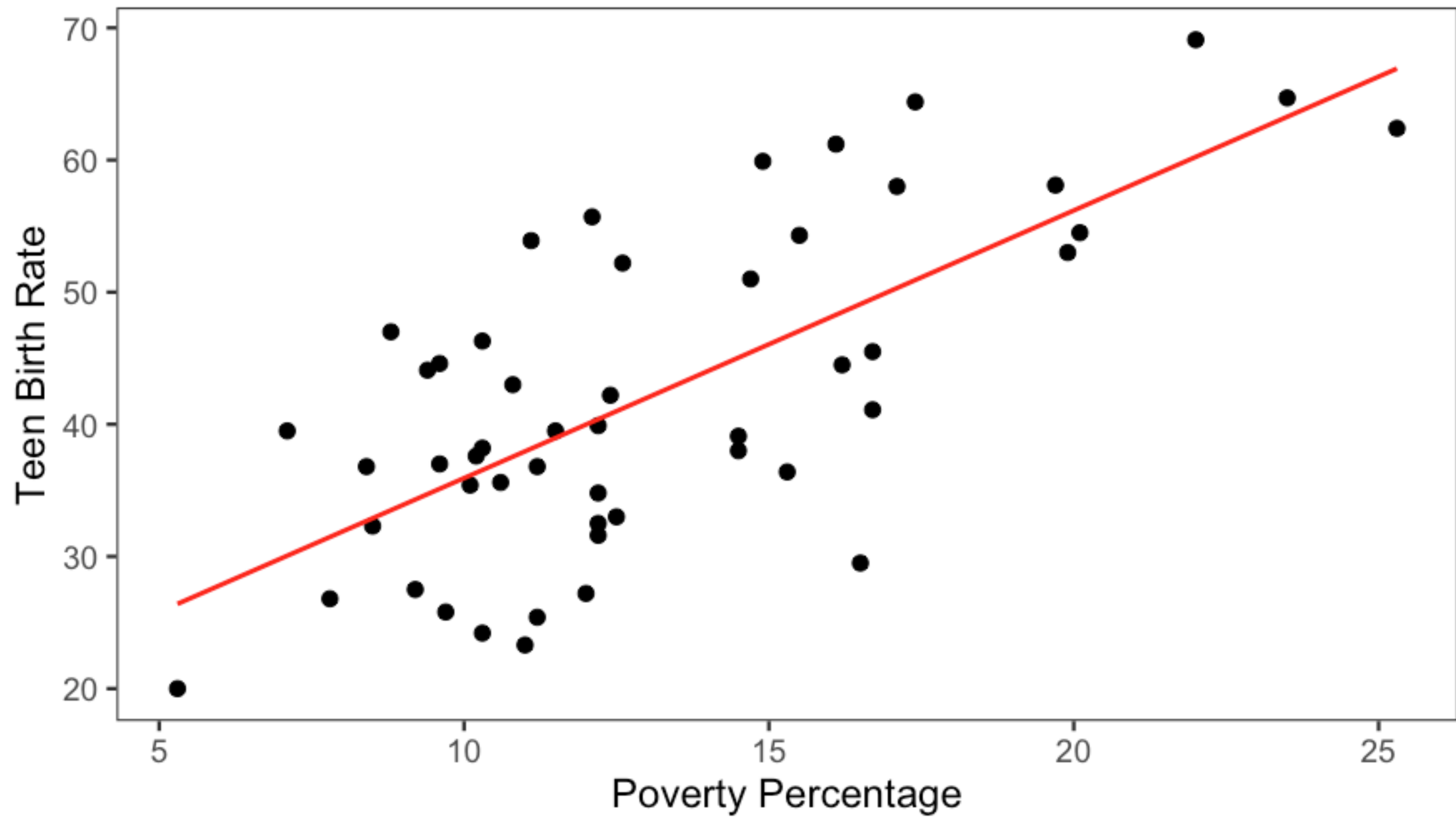
Effect size ( $\beta$ ) can  
be estimated using  
the slope of the line

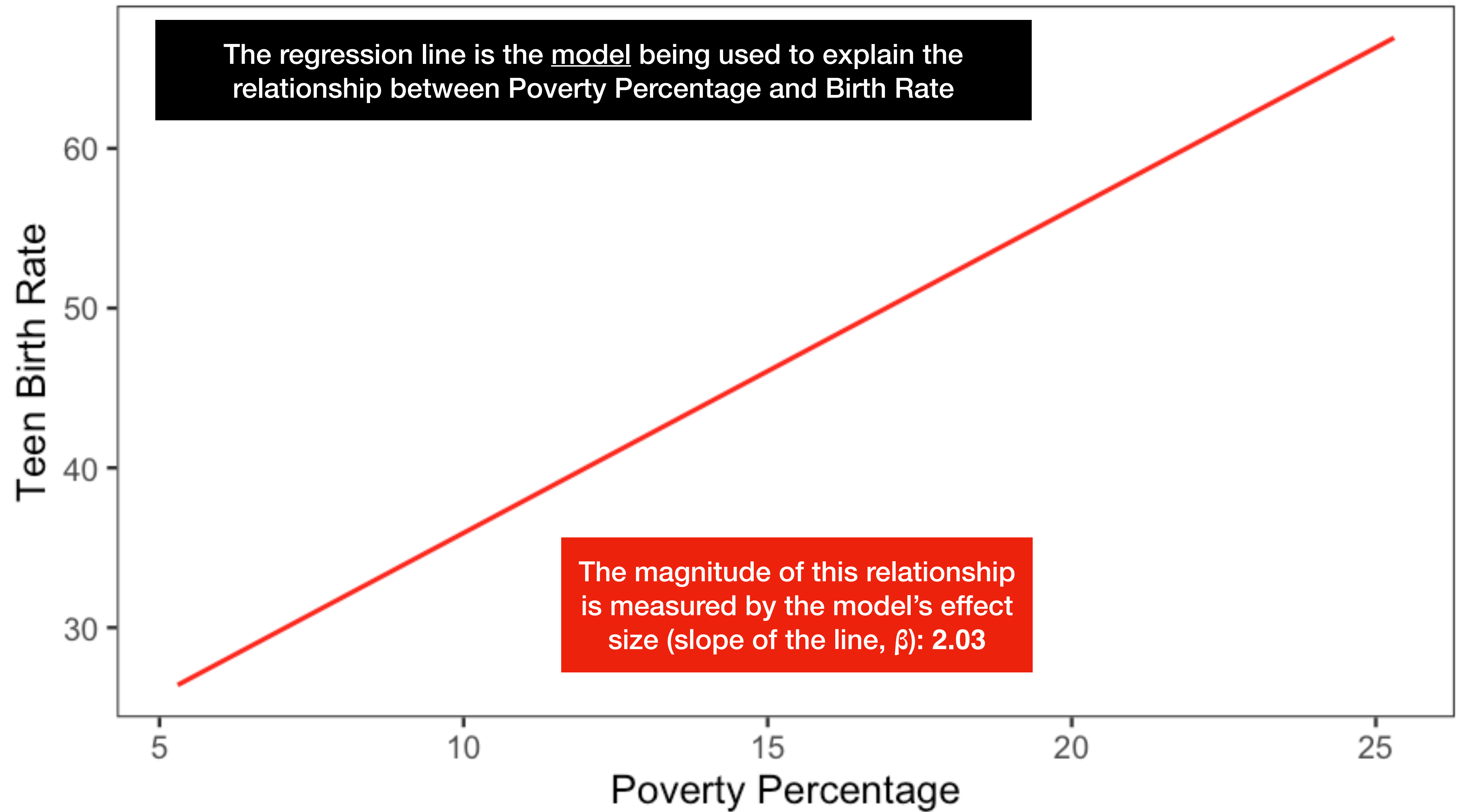


Effect size ( $\beta$ ) can be estimated using the slope of the line

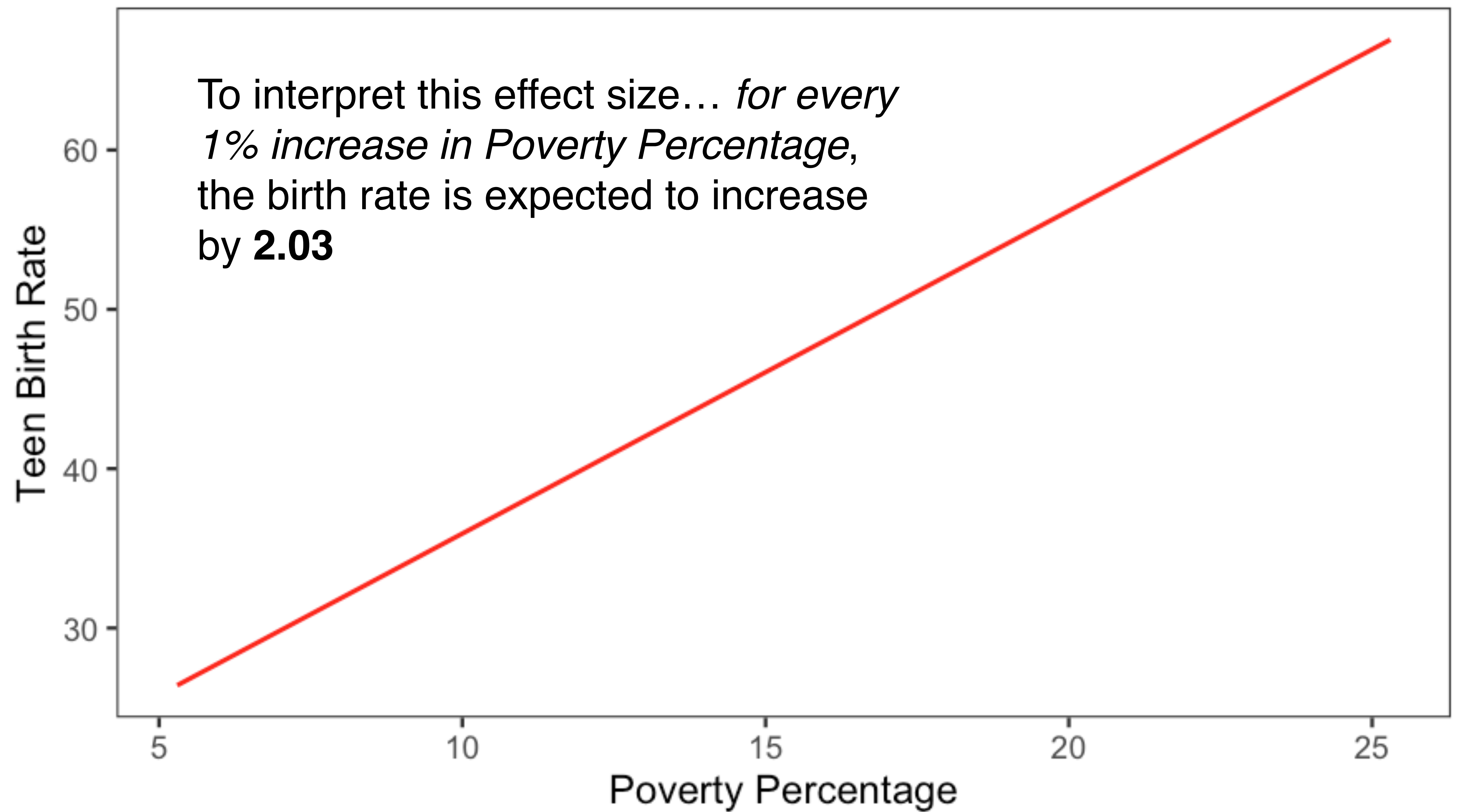












...but *how confident* are we in that estimate of the effect size?

For that...we need to look at the standard error (SE) on the estimate of slope

Teen Birth Rate

Formula

$$SE = \frac{\sigma}{\sqrt{n}}$$

SE = standard error of the sample

$\sigma$  = sample standard deviation

$n$  = number of samples

Poverty Percentage

60

50

40

30

5

10

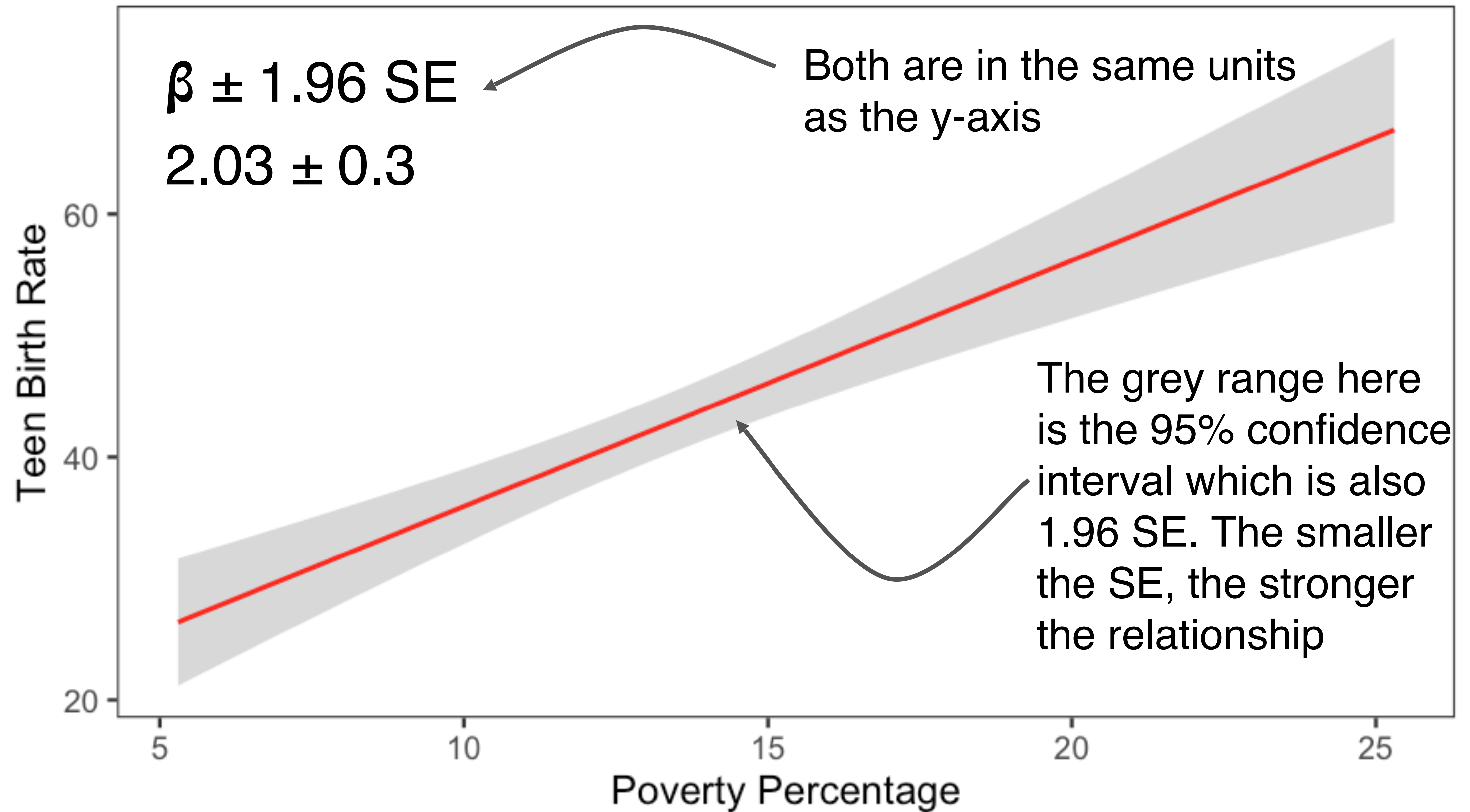
15

20

25

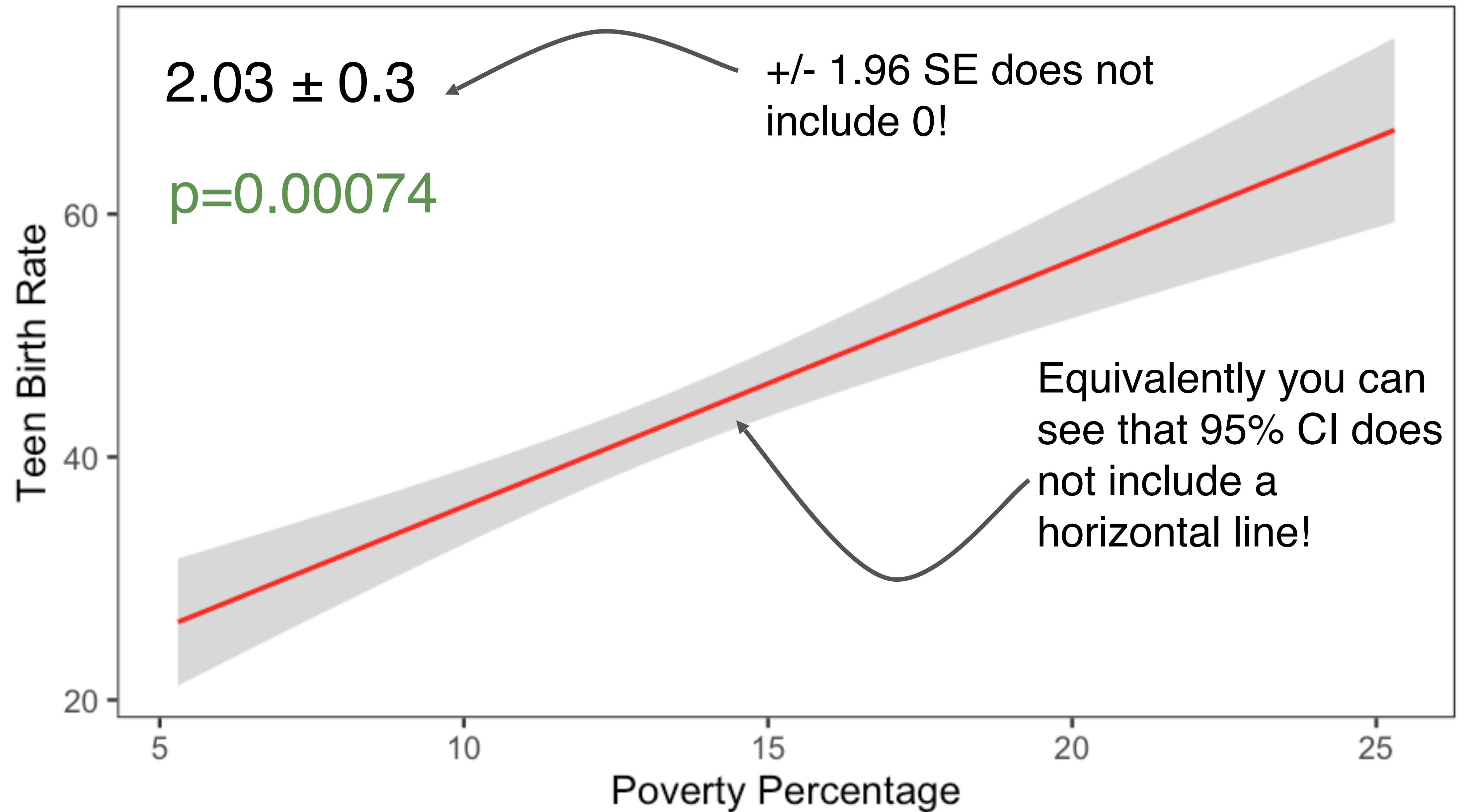
**Confidence interval** : a range of values calculated from a sample statistic, such that there is a specified probability that the value of the true value of the population (parameter) lies within it.





**p-value** : the probability of getting the observed results (or results more extreme) by chance alone under the null hypothesis

Is a function of effect size ( $\beta$ ) and SE!

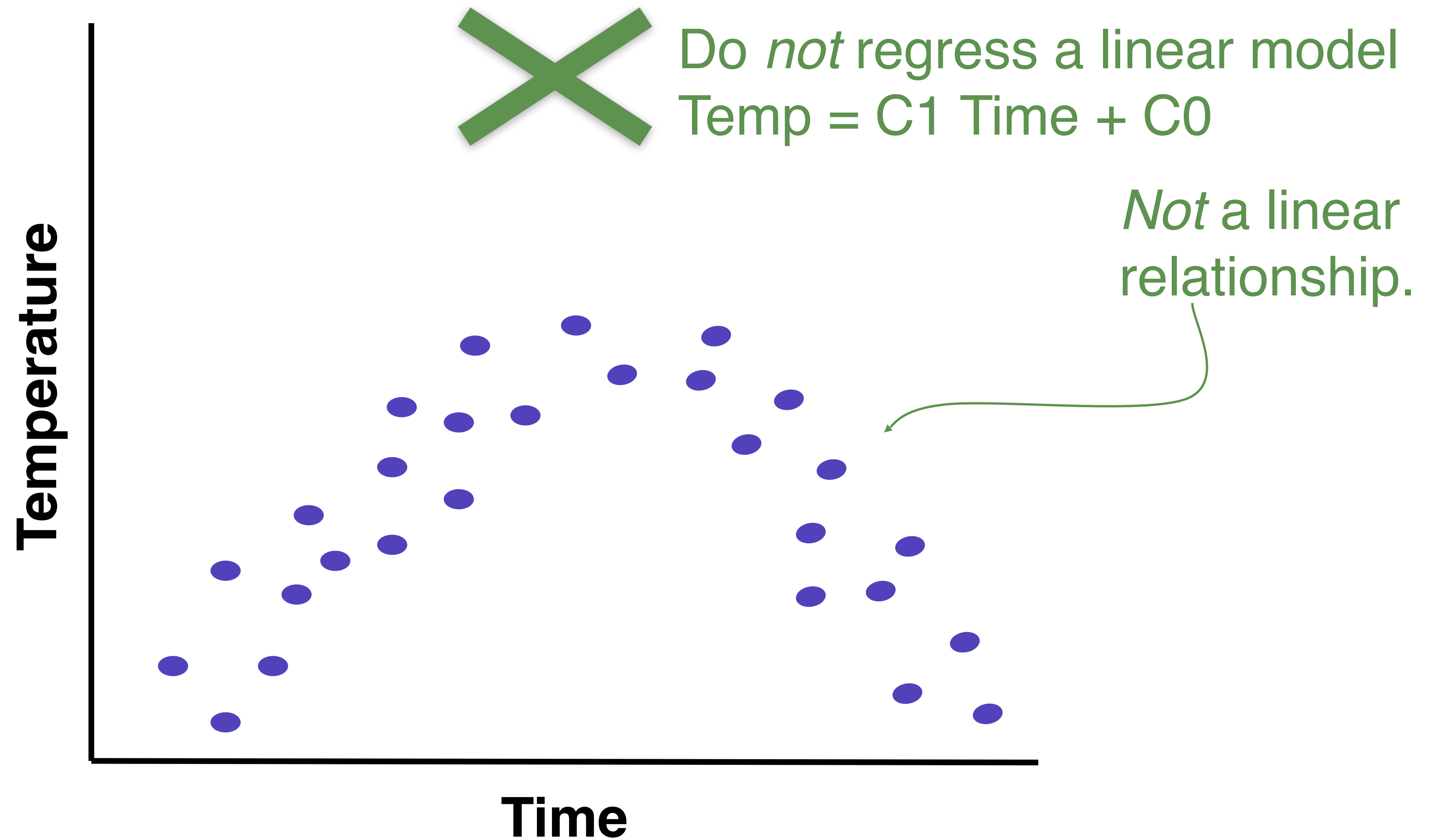




# Assumptions of linear regression

1. Linear relationship
2. ~~Multivariate normality~~ Normal residuals
3. No multicollinearity
4. No autocorrelation
5. Homoskedasticity

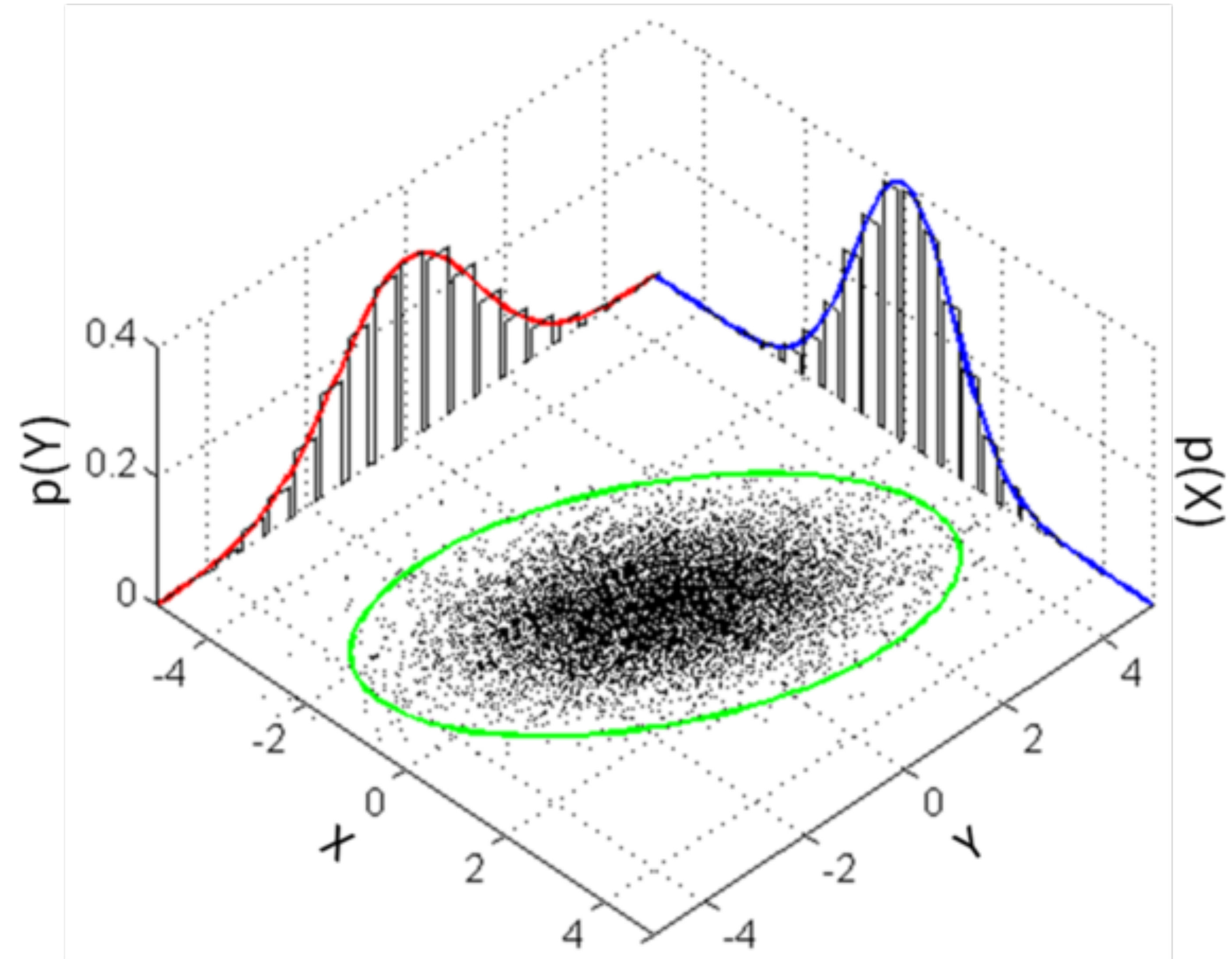




But you can use OLS to regress  
 $\text{Temp} = C2 \text{ Time}^2 + C1 \text{ Time} + C0$



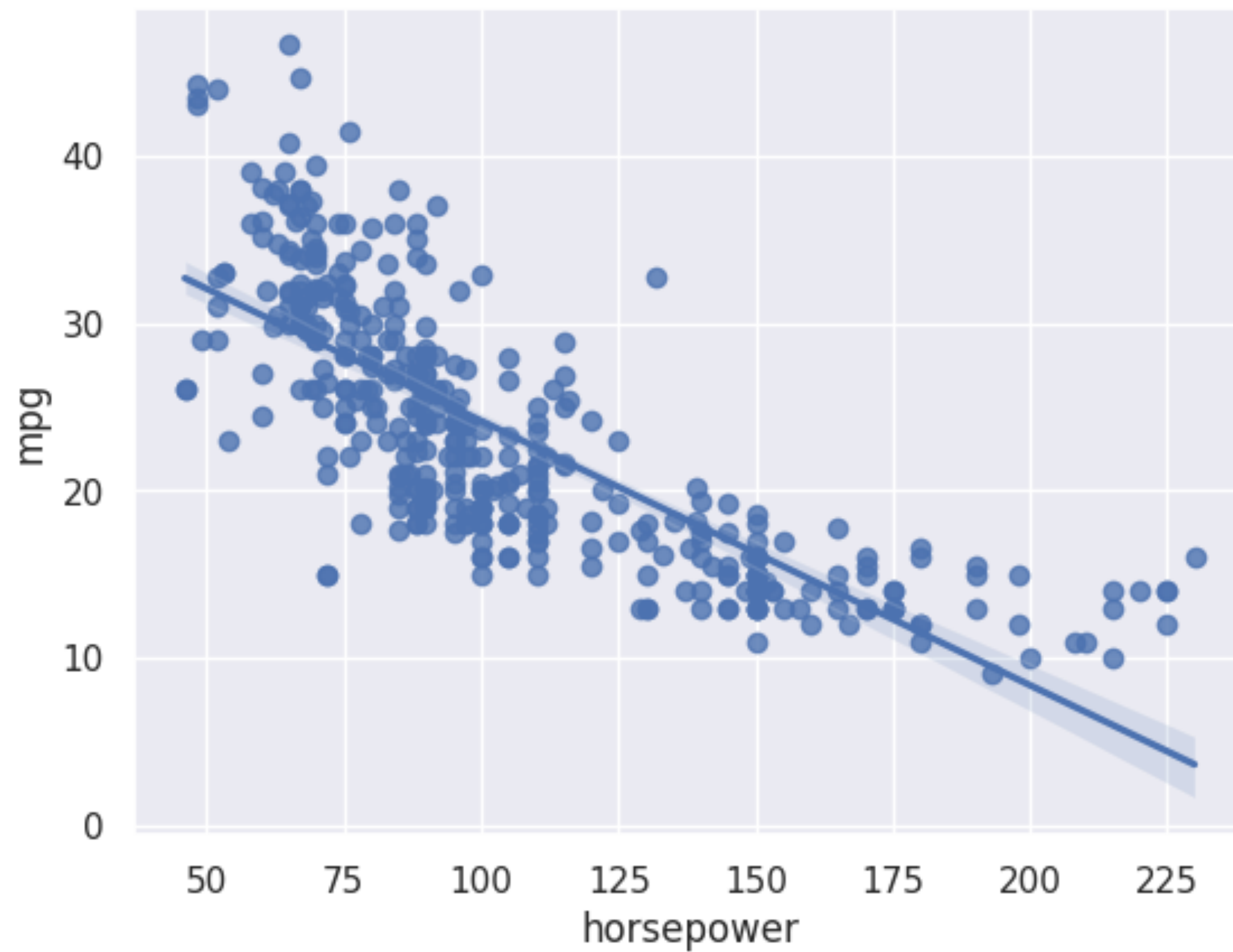
A multivariate  
normal probability  
distribution (joint  
normal)



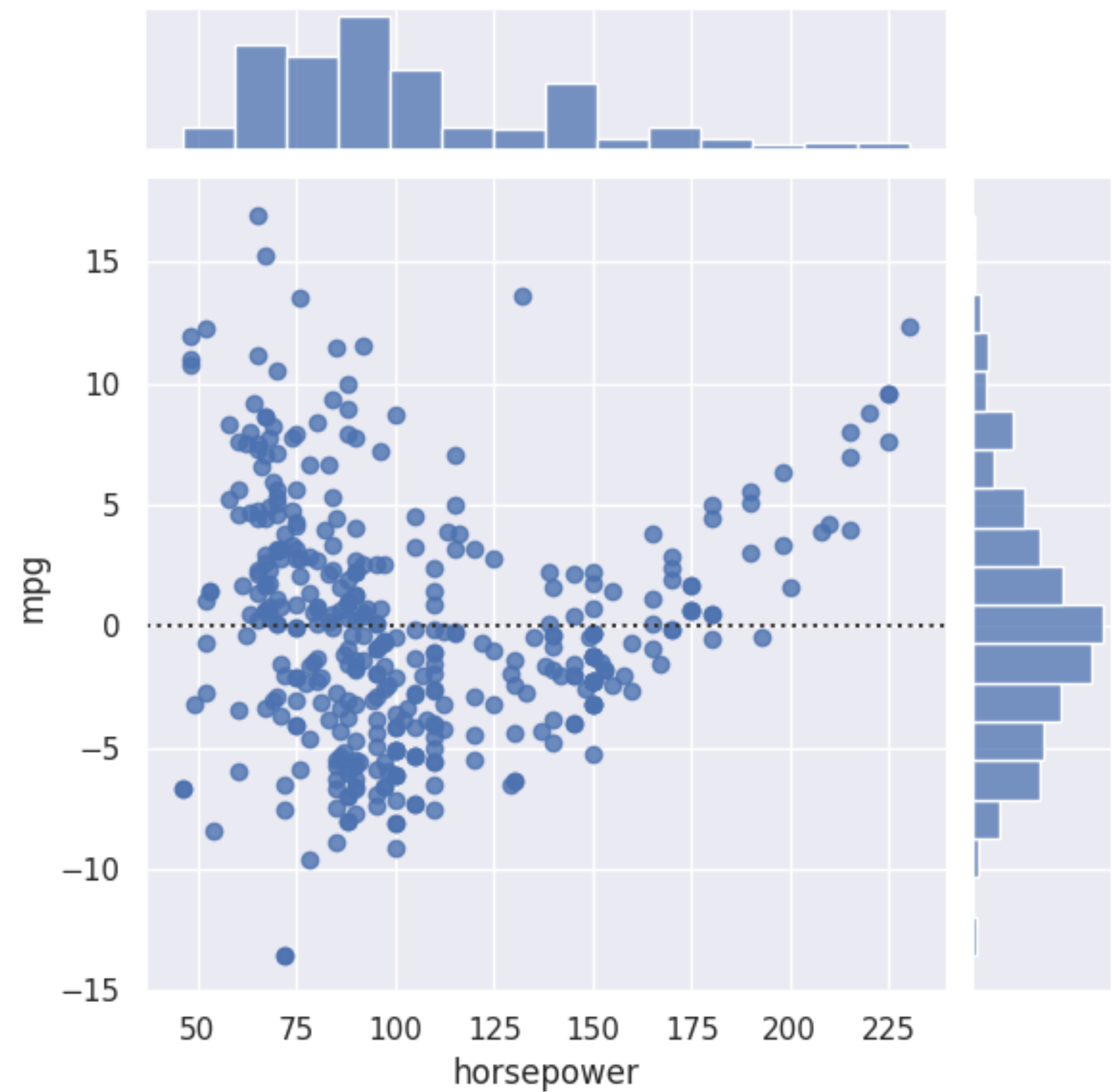
```
mpg = sns.load_dataset("mpg")
```

```
sns.regplot(data=mpg, x="horsepower", y="mpg")  
sns.jointplot(data=mpg, x="horsepower", y="mpg", kind='resid');
```

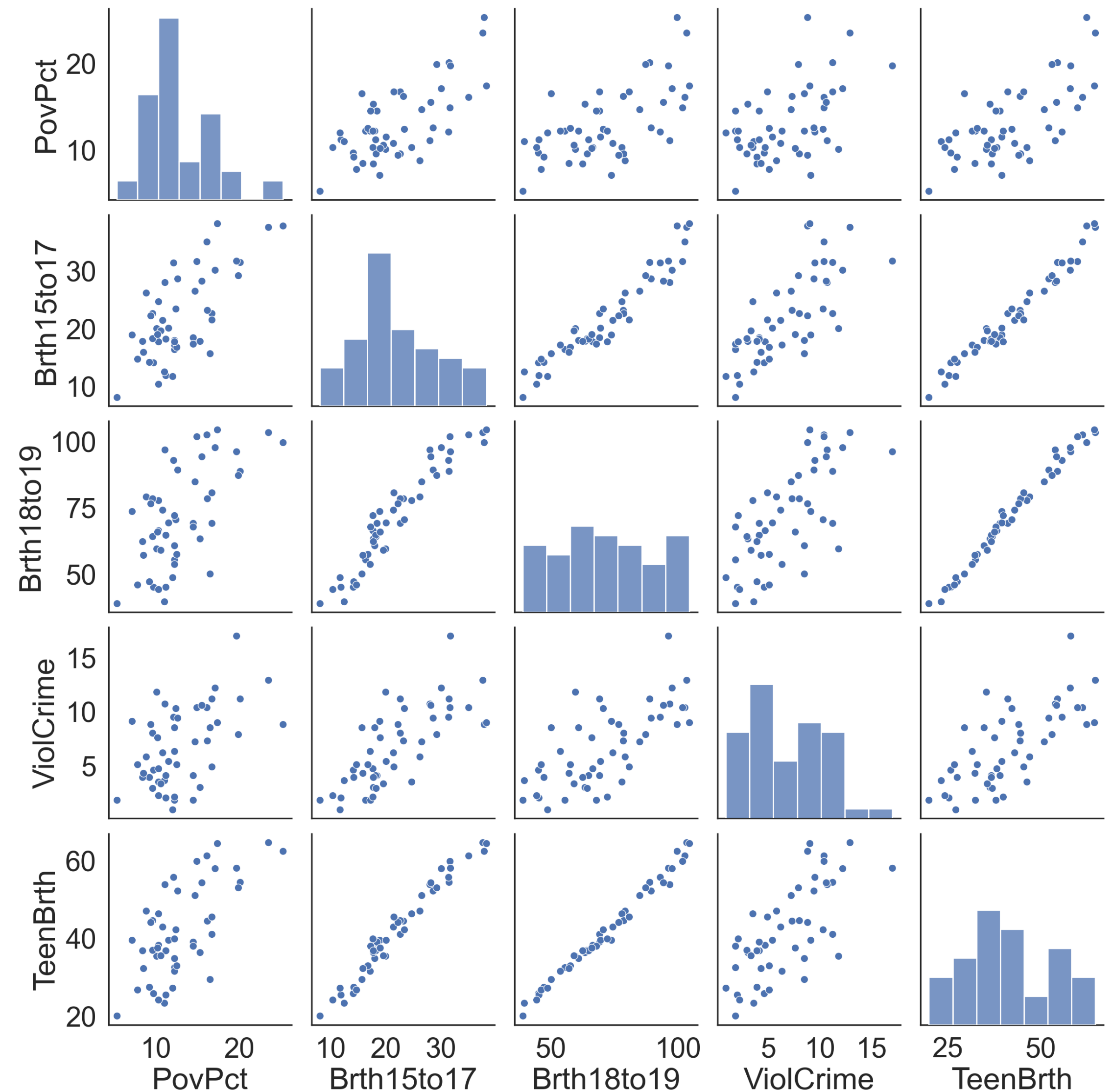
## Regression plot



## Residual plot



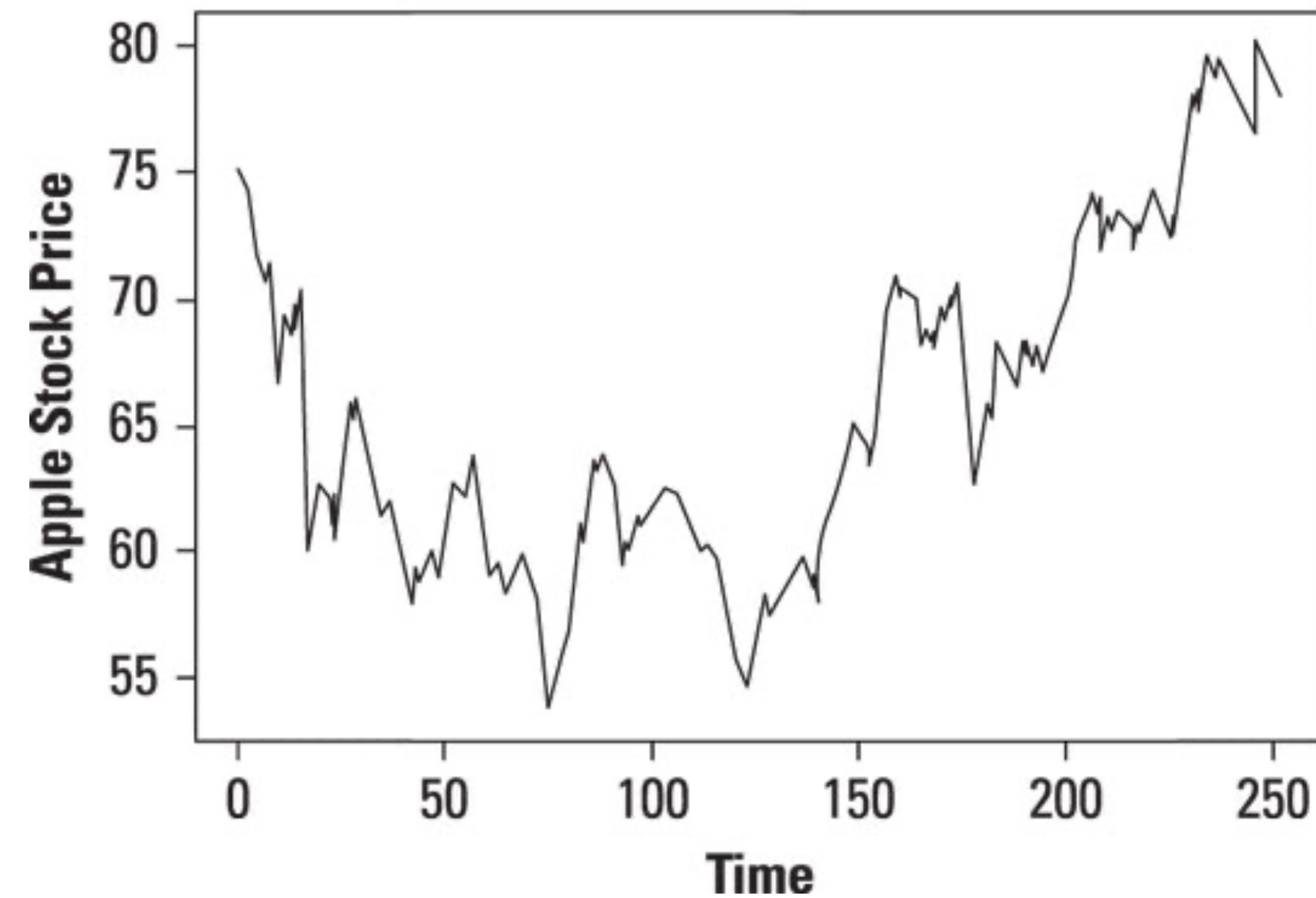
**Multicollinearity** occurs when the independent variables (in multiple linear regression) are too highly correlated with each other.



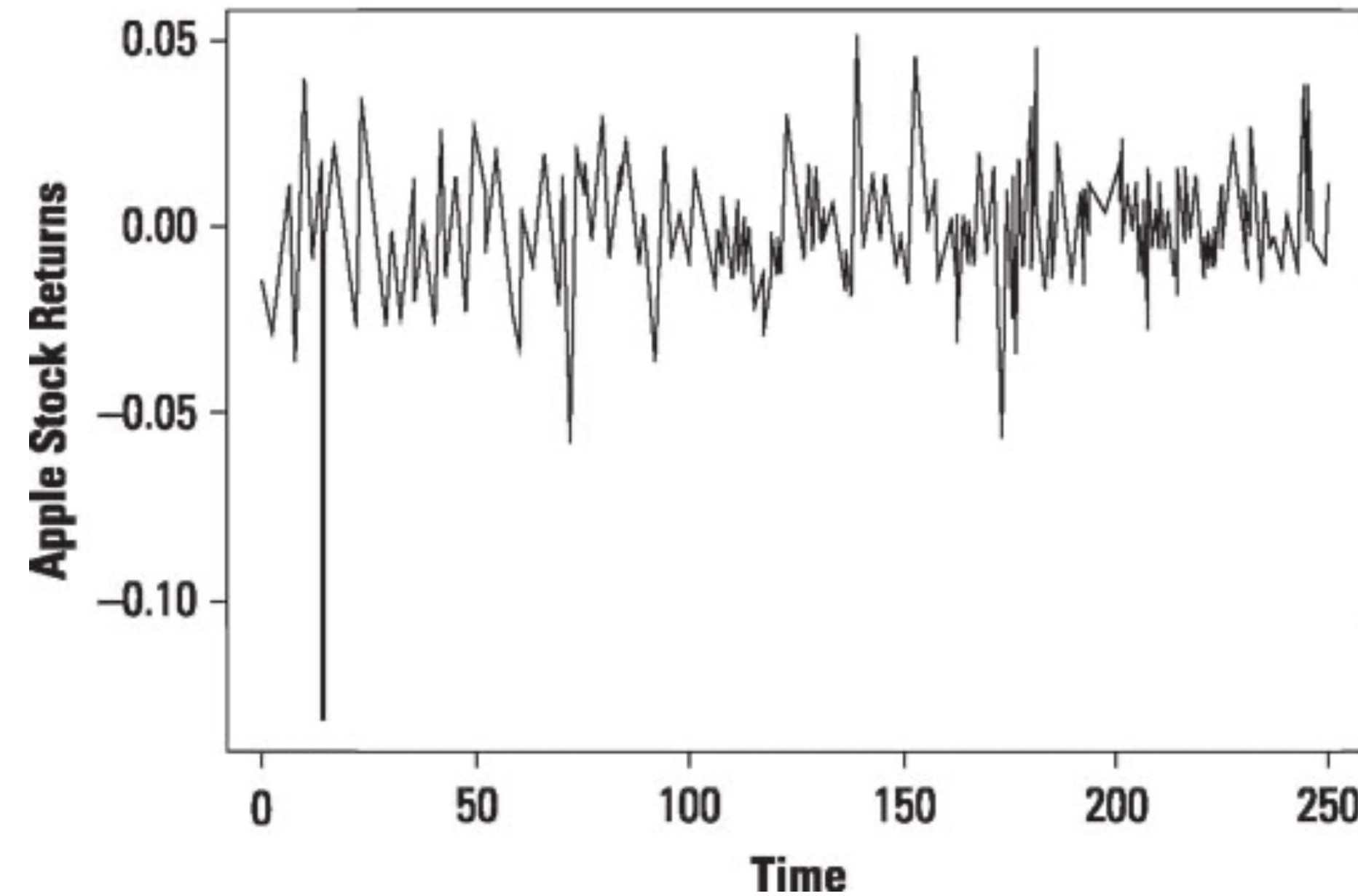


Daily returns are  
 $\ln(\text{price}_t / \text{price}_{t-1})$

Time Series Plot of Apple Stock Prices



Time Series Plot of Apple Stock Returns

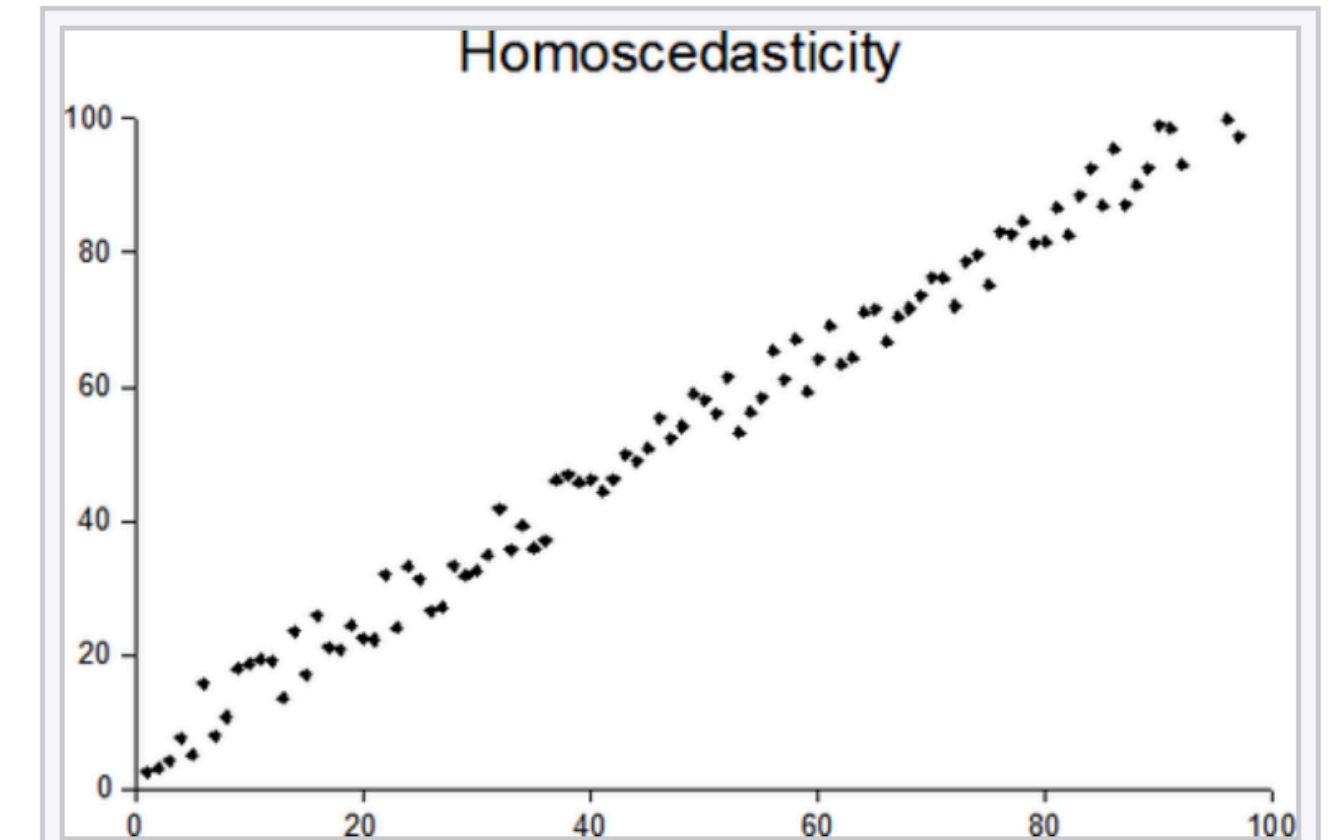


**Autocorrelation** occurs when the observations are *not* independent of one another (i.e. stock prices)

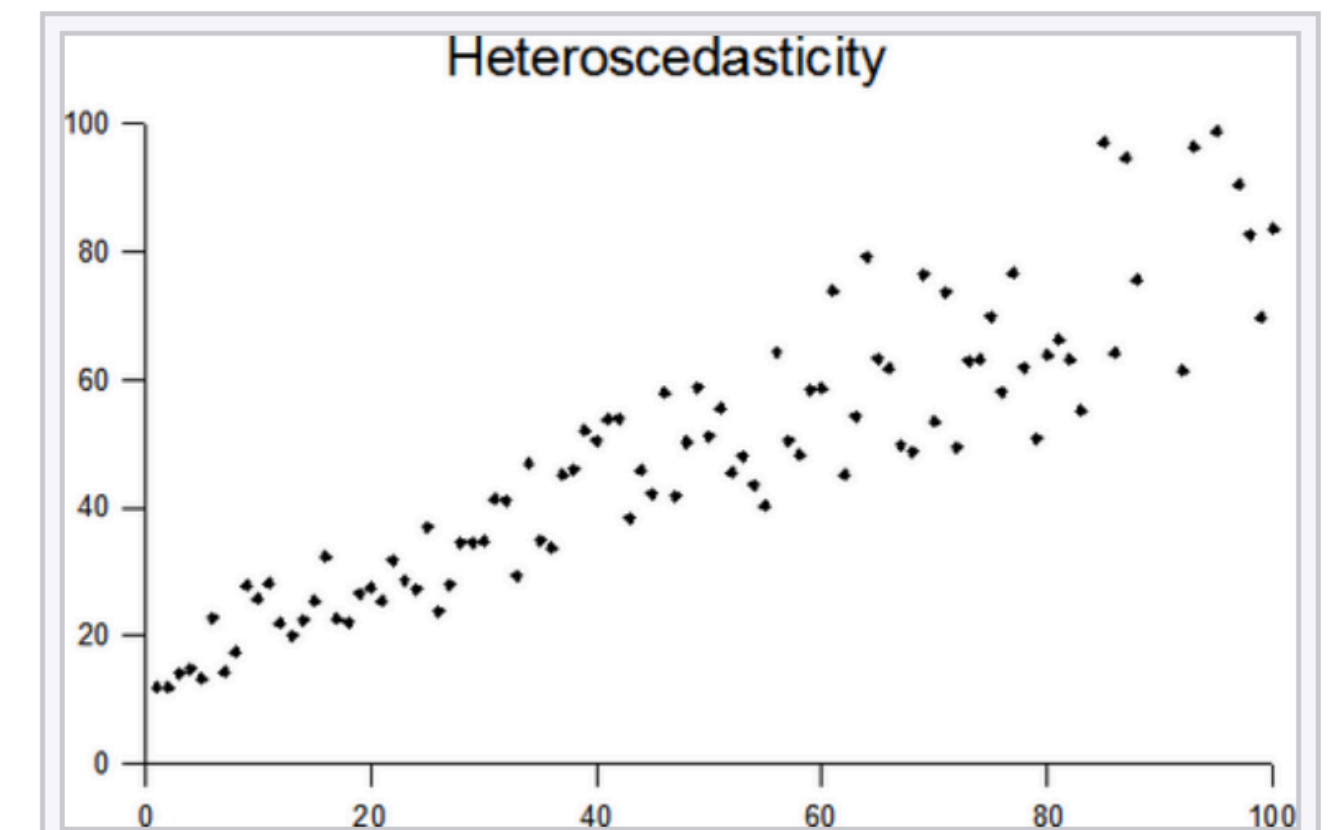
Heteroscedasticity occurs when the variance in the dependent variable is the same regardless of the value of the independent variable



Do *not* use  
OLS  
regression



Plot with random data showing homoscedasticity: at each value of  $x$ , the  $y$ -value of the dots has about the same **variance**.



Plot with random data showing heteroscedasticity: The variance of the  $y$ -values of the dots increases with increasing values of  $x$ .

# Modeling spatial or density or inter-event timing data: Poisson Point Process

$$\lambda(i) = e^{\alpha + \beta Z(i)}$$

$\lambda(i)$  is the modeled density at location  $i$

$e^\alpha$  is the base intensity when the covariate is *zero*

$e^\beta$  is the multiplier by which the intensity increases (or decreases) for each 1 unit increase in the covariate



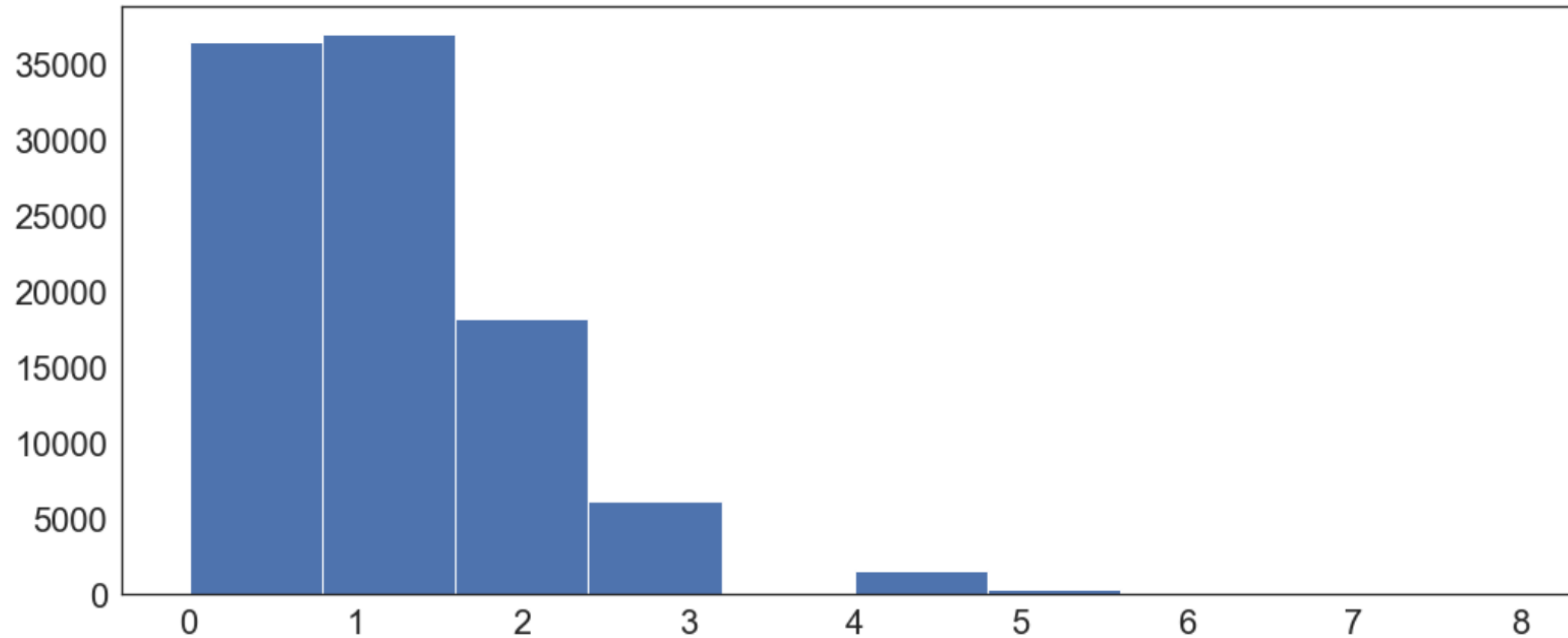
## Poisson Distribution

The Poisson Distribution models events in fixed intervals of time, given a known average rate (and independent occurrences).

In [55]:

Slide Type Fragment ▾

```
dat = poisson.rvs(mu=1, size=100000)
plt.hist(dat);
```



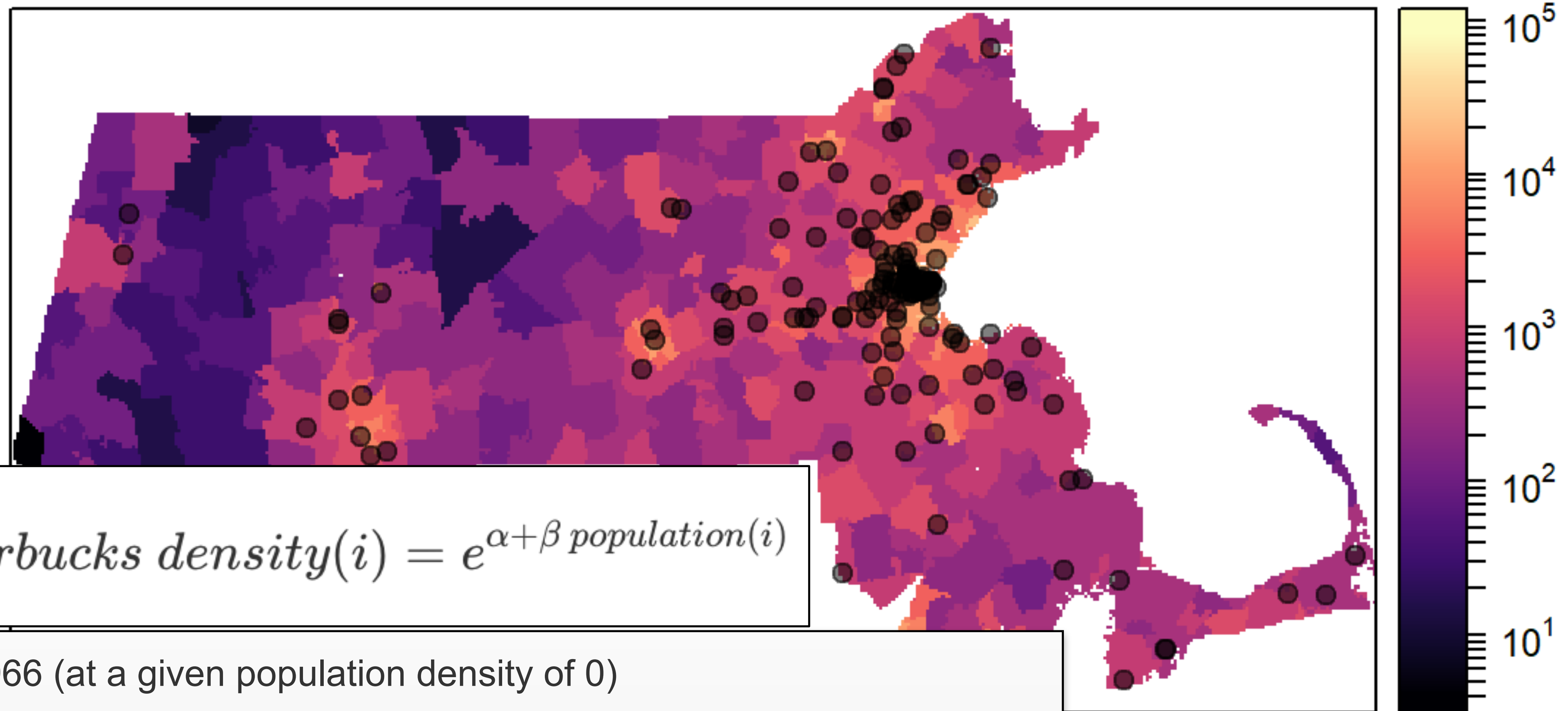
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The **number of visitors a fast food drive-through gets each minute** follows a Poisson distribution. In this case, maybe the average is 3, but there's some variability around that number.

A Poisson distribution can help calculate the probability of various events related to customers going through the drive-through at a restaurant. It will predict lulls (0 customers) and flurry of activity (5+ customers), allowing staff to plan and schedule more precisely.



## Location of Starbucks relative to population density in MA



$$\text{Starbucks density}(i) = e^{\alpha + \beta \text{ population}(i)}$$

$\alpha = -18.966$  (at a given population density of 0)

$e^{-18.966} = 5.80 \times 10^{-9}$  cafes per square meter

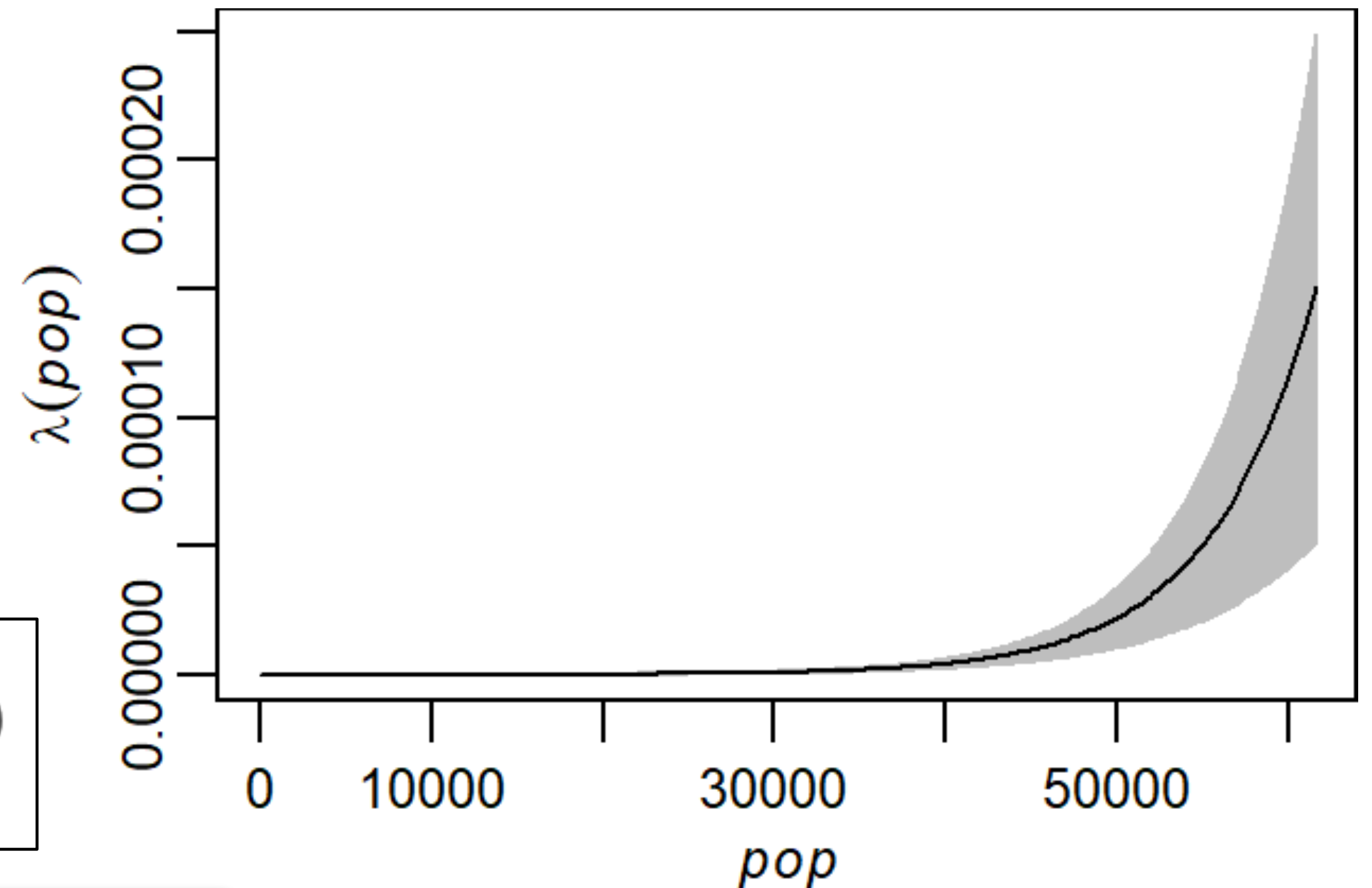
$\beta = 0.00017$ ;  $e^{0.00017}$  or 1.00017



# Location of Starbucks relative to population density in MA

Null hypothesis is CSR  
which means  $\beta = 0$

$$\text{Starbucks density}(i) = e^{\alpha + \beta \text{ population}(i)}$$

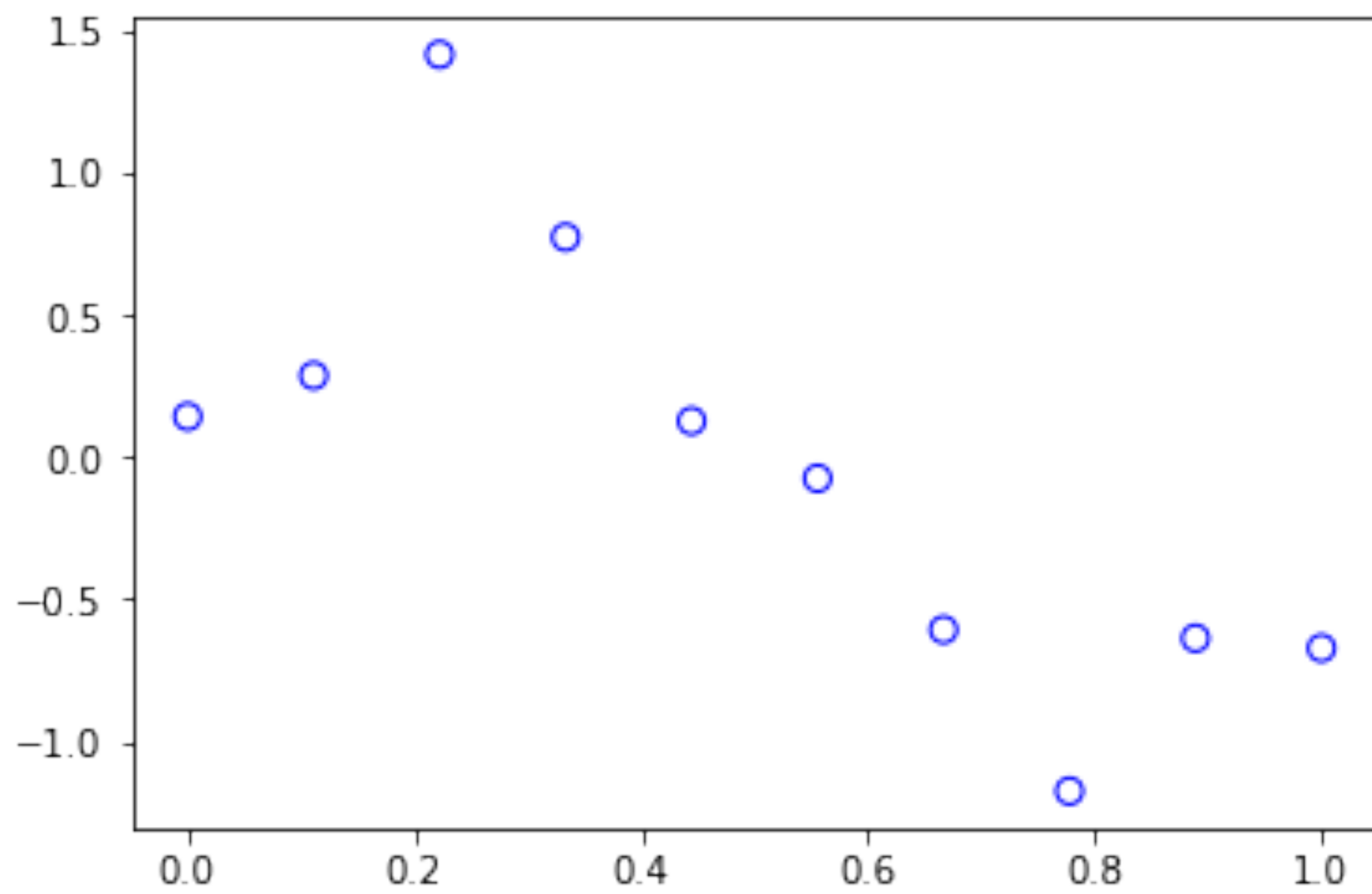


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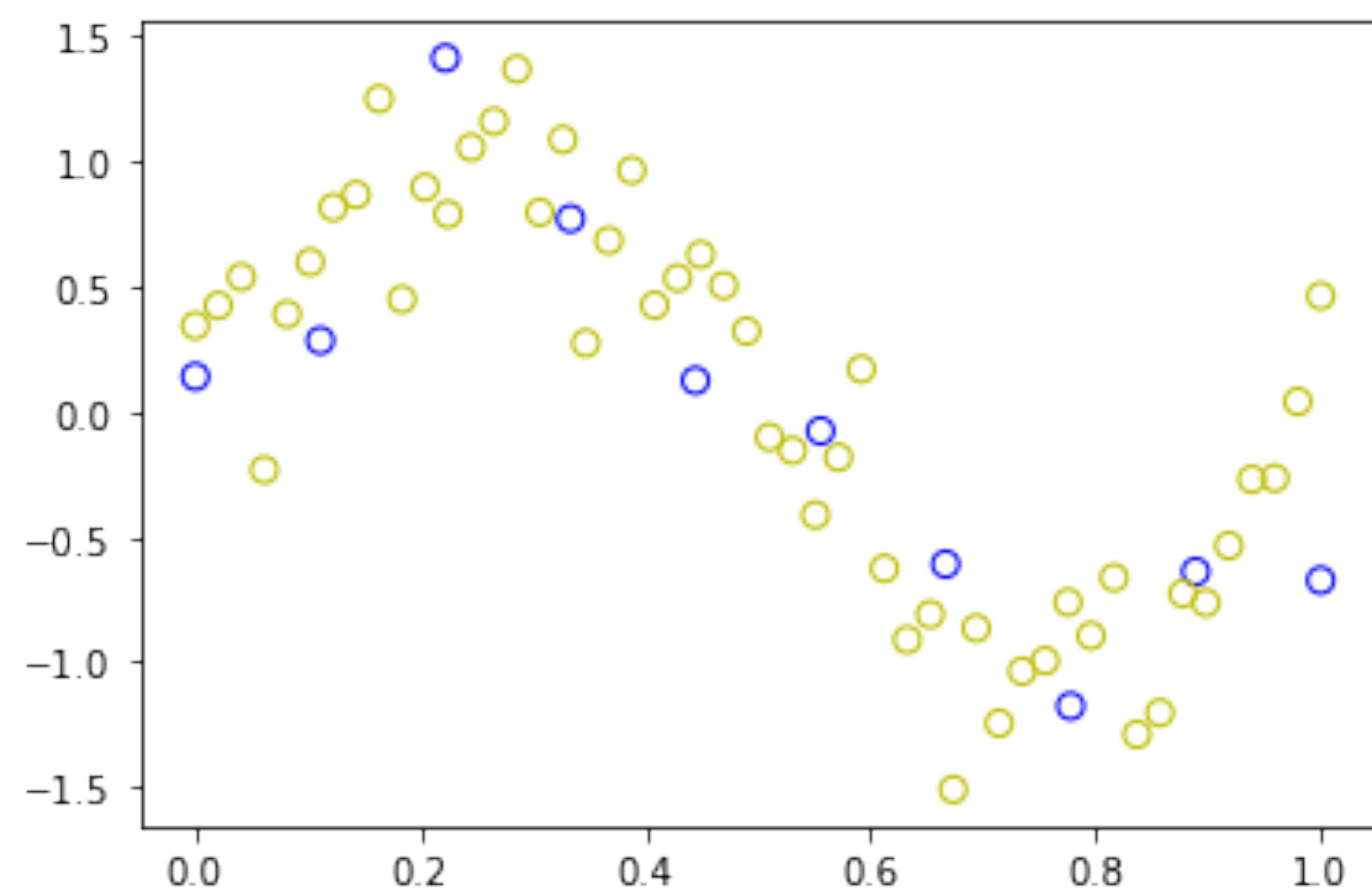
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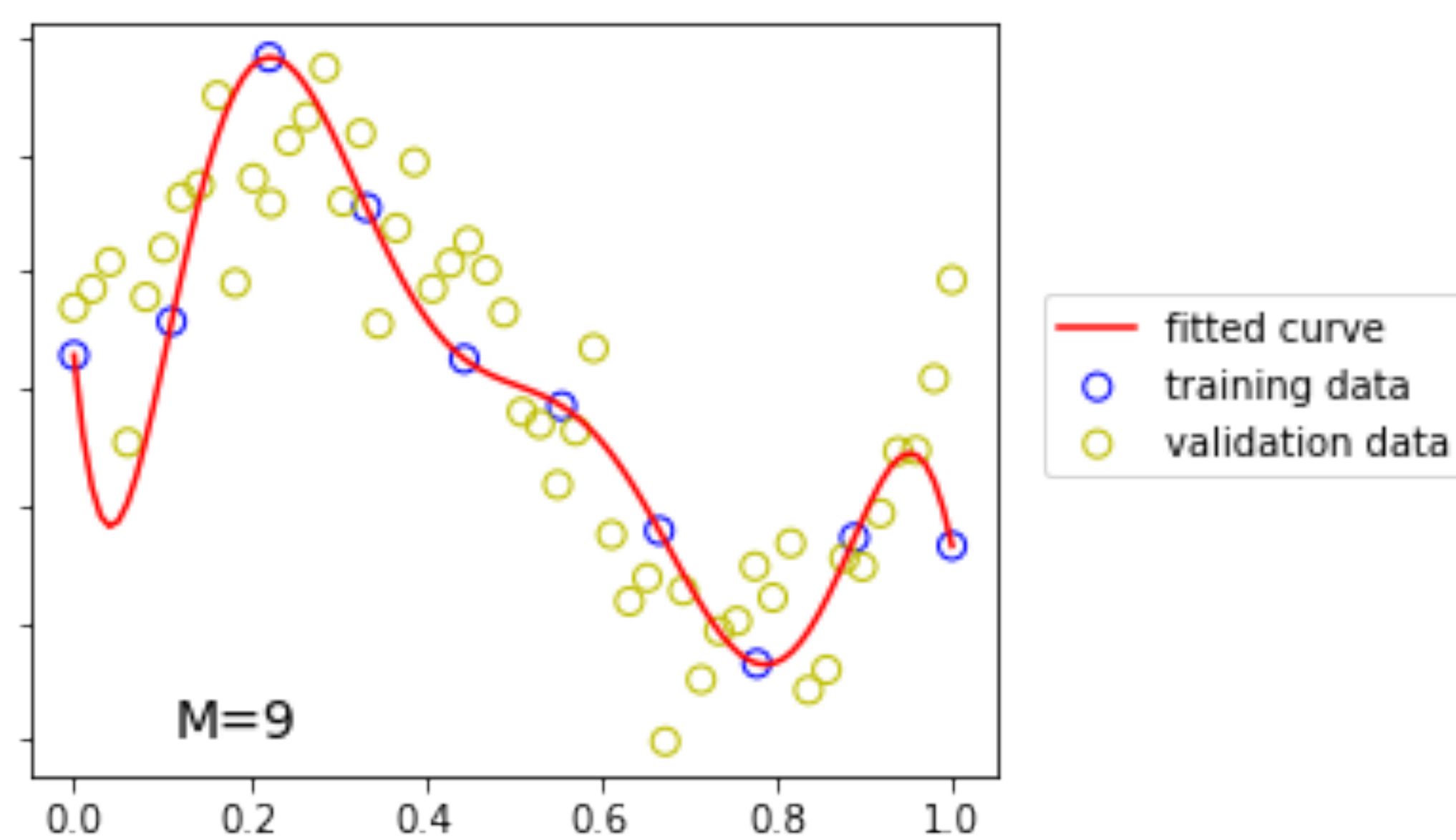
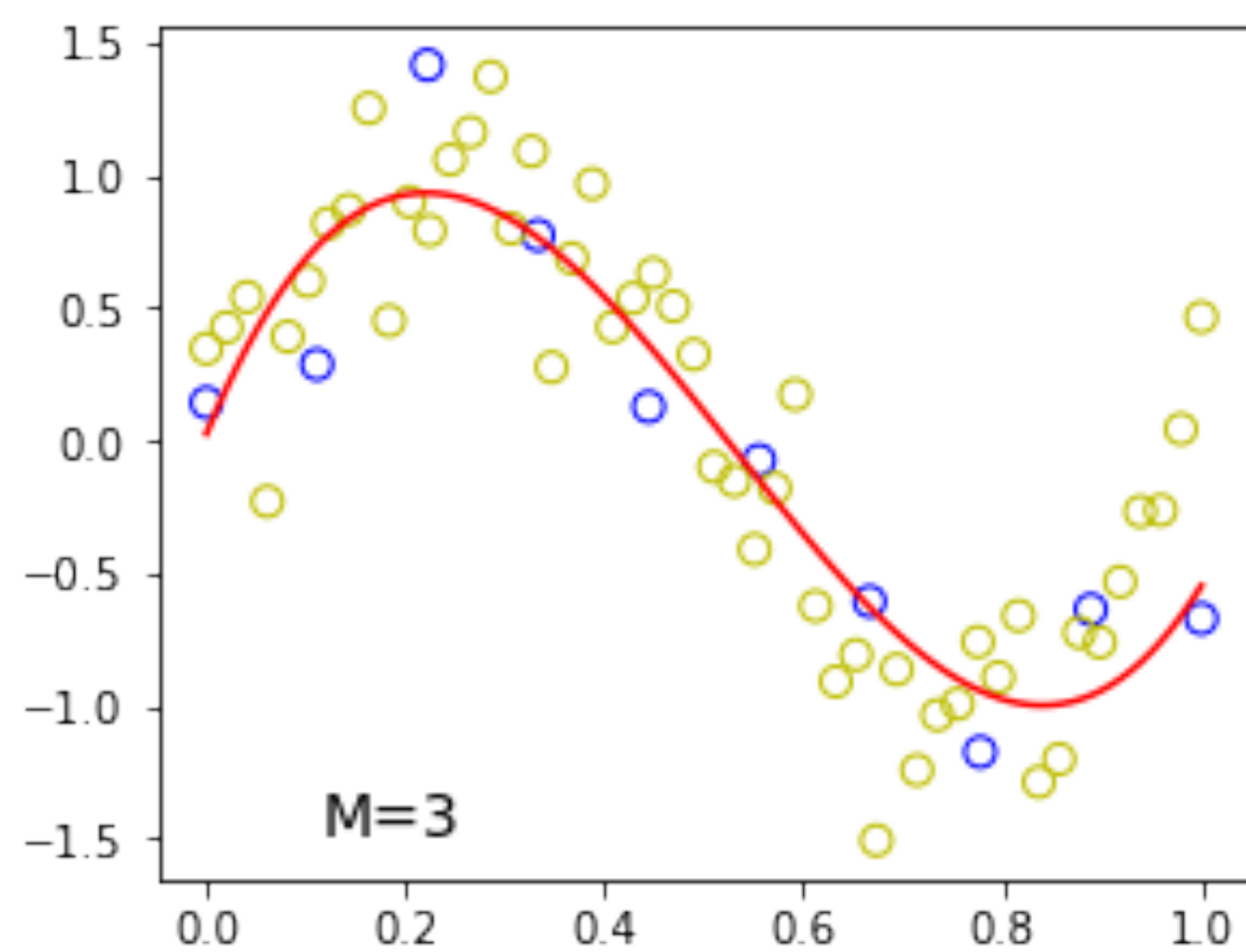
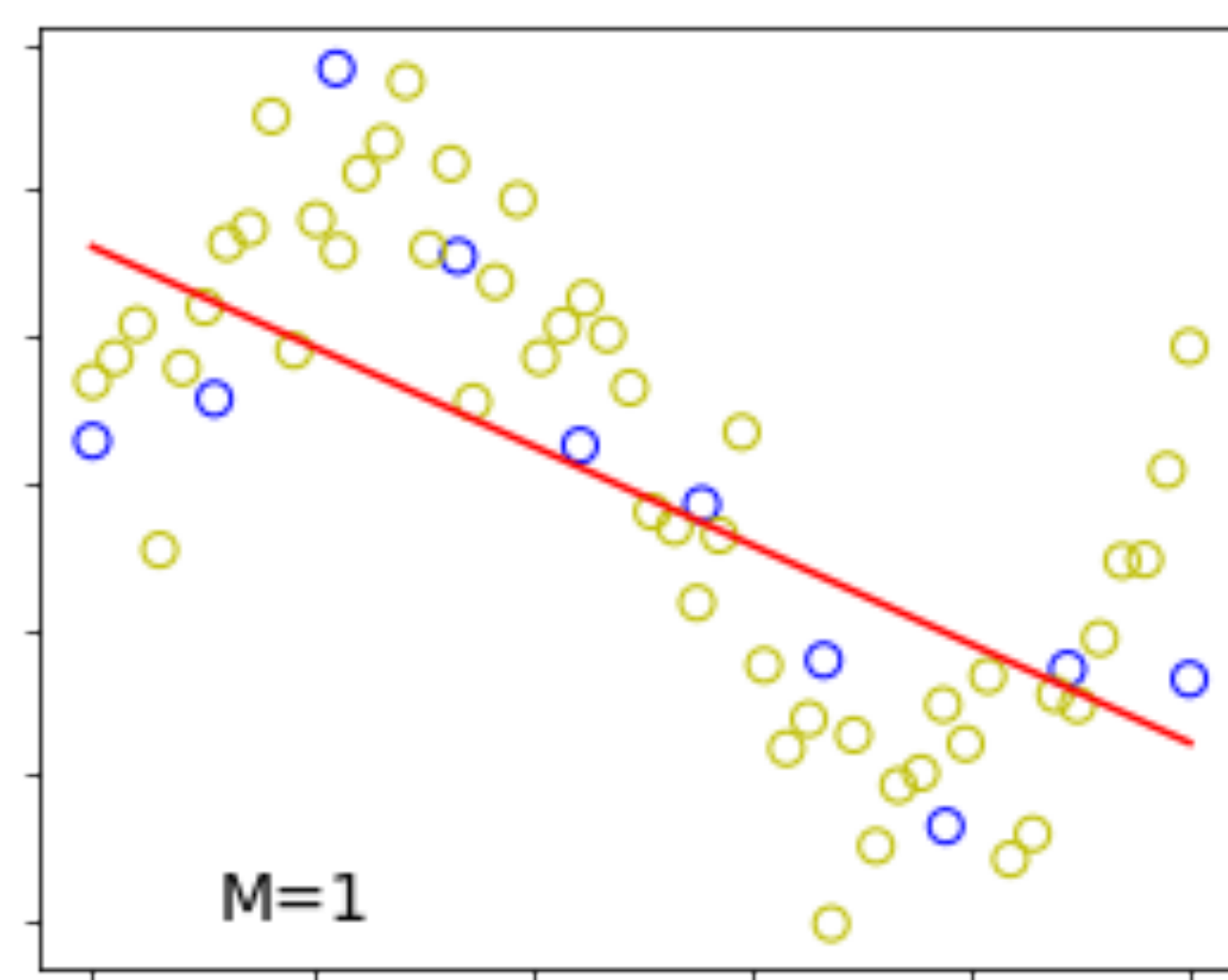
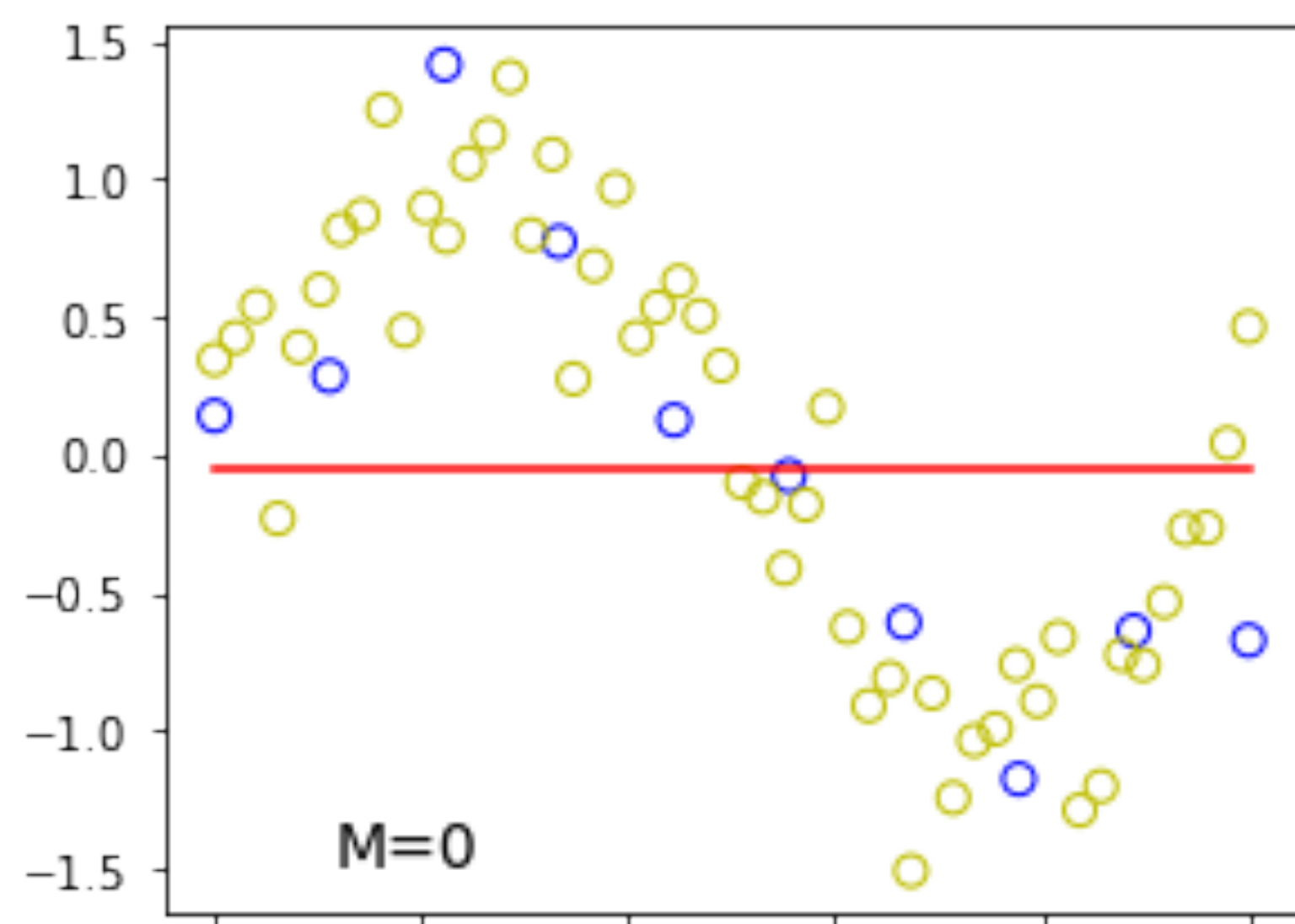
$\beta = 0.00017$ ;  $e^{0.00017}$  or 1.00017

Model selection problems









# Overfitting vs underfitting

- Models in statistical inference have this form
$$y = f(x) + \mathcal{N}(\mu, \sigma)$$
  - Deterministic relationship plus some noise
- Overfit: the model is TOO COMPLEX, it has learned about the noise
  - If you have zero error on training data you are overfit by definition
- Underfit: the model is TOO SIMPLE to capture the deterministic relationship
  - For example using  $M=1$  on the data which are sinusoidal



## 2.3 Parsimony

Since all models are wrong the scientist cannot obtain a “correct” one by excessive elaboration. On the contrary following William of Occam he should seek an economical description of natural phenomena. Just as the ability to devise simple but evocative models is the signature of the great scientist so overelaboration and overparameterization is often the mark of mediocrity.

## 2.4 Worrying Selectively

Since all models are wrong the scientist must be alert to what is importantly wrong. It is inappropriate to be concerned about mice when there are tigers abroad.



# Goodness of fit + Parsimony

## How to measure model fitness

- Adjusted R-squared  
AKA % variance explained
- Higher is better
- Range [0,1]

$$R_{adj}^2 = 1 - \frac{(1 - R^2)(n - 1)}{n - p - 1}$$

where :

$R^2$  = R - squared

$n$  = number of samples/rows in the data set

$p$  = number of predictors/features

- Bayesian Information Criterion
- Lower is better
- Range [-inf, +inf],  
negative values usually indicated a good fit

$$\text{BIC} = -2 \cdot \ln(L) + k \cdot \ln(n)$$

Where:

- $\ln(L)$ : The natural logarithm of the **likelihood function**, which evaluates how well the model fits the data.
- $k$ : The **number of parameters** in the model, representing complexity.
- $n$ : The **number of observations** or data points in the dataset.

# Two ways to do model selection

- Out of sample
  - ML method
  - Training set vs validation set
  - Select model by performance metric on validation set
- In sample
  - Statisticians method
  - Use a selection criteria with a metric of the form:  
$$M = \text{goodness of fit} + \text{penalty for a complex model}$$

# In sample model selection: Best subset

An example dataset: [TeenBrth[dependent], ViolCrime, PovPct, AvgIncome]

- Try every possible combination of independent variables, select the best model according to BIC or other metric
  - TeenBrth ~ ViolCrime
  - TeenBrth ~ PovPct
  - TeenBrth ~ AvgIncome
  - TeenBrth ~ ViolCrime + PovPct
  - TeenBrth ~ AvgIncome + PovPct
  - TeenBrth ~ ViolCrime + AvgIncome
  - TeenBrth ~ ViolCrime + PovPct + AvgIncome
- When you have too many variables (best subset fits  $2^n$  models) use forward selection instead (see next slide)



# Forward model selection

## In-sample or Train/Validation, either works

- A step-by-step algorithm for feature selection that starts with an empty model and iteratively adds the most helpful variable—based on penalized metrics like Adjusted R-squared, AIC, or BIC—until no further improvement is achieved.
- Computationally efficient alternative to best subset selection that reduces overfitting by excluding irrelevant variables, though it may miss the optimal model.
- Procedure: Begins with a "null" model (intercept only). In each step, it tests all variables not currently in the model and adds the one that best improves the model fit.
- Stopping Criteria: The process stops when no remaining variables improve the model fit (penalized metric), or the added variable is not significant by a predetermined threshold, such as a p-value  $> 0.05$ .
- Advantages: It is highly efficient for high-dimensional data, helps prevent overfitting, and is easy to interpret.
- Disadvantages: It is a "greedy" algorithm, meaning it makes the best choice at each step, which might not lead to the overall best model. It can also struggle with highly correlated predictors.
- Forward Selection vs. Other Methods
  - vs. Backward Elimination: Starts with all variables and removes them, whereas forward starts with none.
  - vs. Best Subset: Forward selection is computationally faster because it does not test every possible combination of variables.

# In sample model selection: Forward Selection

An example dataset: [TeenBrth[dependent], ViolCrime, PovPct, AvgIncome]

- Start with the null model. Try every variable you don't have yet. Keep the best model according to BIC or other metric. Then go again adding new variables until the selection metric does not improve by adding new variables.
  - TeenBrth ~ ViolCrime BIC=-1100
  - TeenBrth ~ PovPct BIC=-1000
  - TeenBrth ~ AvgIncome BIC=-900
  - TeenBrth ~ ViolCrime + PovPct BIC=-1350
  - TeenBrth ~ ViolCrime + AvgIncome BIC=-1200
  - TeenBrth ~ ViolCrime + PovPct + AvgIncome BIC=-1310
- Fits only  $n(n+1)/2$  models vs best fit which is  $2^n$

# Model selection in sample

